Derivation of the Long-Distance Hyperfine–Zeeman Relation in an RKB Basis

Reading guide: every section states, in plain words, what we are trying to get before showing the result. All algebra is worked out step by step in the clickable appendices.

Central result — the one-sentence version

What we wanted: an operator that tells us how the Dirac Hamiltonian responds to the magnetic moment of a nucleus K that sits *far away* from the electron density we are actually computing with (paramagnetic centre at O). **What we found:** to leading order in $|\boldsymbol{r}_O|/R$, this distant-nucleus hyperfine operator is *literally the ordinary Zeeman operator*, geometrically reshaped by the point-dipole tensor $T(\boldsymbol{R})$, plus a piece that is mathematically forced to vanish once we take an expectation value in a real (stationary) electronic state:

$$\boxed{\left(\frac{\partial h^D}{\partial M_{K,l}}\right)^{(0)}_{\mathrm{LD}} \simeq \alpha^2 \sum_m T_{ml}(\boldsymbol{R}) \left(\frac{\partial h^D}{\partial B_m}\right)^{(0)} + \mathrm{i}\,[h^{D,(0)}, f_l].} \tag{1}$$

Here

$$T_{ml}(\boldsymbol{R}) = \frac{3R_m R_l - R^2\delta_{ml}}{R^5} \tag{2}$$

is the point-dipole tensor. The first term is the complete Zeeman operator, redirected by T_{ml}; the second term is a gauge commutator that drops out for any physical (stationary) state. **The two physical consequences that follow, and that we prove below:**

$$\mathrm{FC}^{\mathrm{LD}} = 0, \qquad \mathrm{SD}^{\mathrm{LD}} = T \cdot \mathrm{SZ}.$$

The Fermi-contact part of the hyperfine coupling has no long-distance counterpart at all; the spin-dipole part turns exactly into the spin-Zeeman part, rescaled by T.

Contents

1 Scope, notation, and order of operations

What this step is for

Fix the notation once ($\boldsymbol{r}_O$, $\boldsymbol{r}_K$, $\boldsymbol{R}$) and pin down the *order* in which we differentiate the Hamiltonian and expand it in $1/R$ — doing these two operations in the wrong order silently changes the answer, so this has to be nailed down before anything else.

The purpose is the same as in the full derivation: first construct the *exact local* hyperfine decomposition without a long-distance approximation, then repeat the same RKB logic after expanding the distant nuclear field around the electronic centre O. The difference in presentation is that the main text keeps the logical steps and final formulas, whereas the line-by-line algebra is moved to the appendices. We use

$$\boldsymbol{r}_O = \boldsymbol{r} - \boldsymbol{R}_O, \qquad \boldsymbol{r}_K = \boldsymbol{r} - \boldsymbol{R}_K = \boldsymbol{R} + \boldsymbol{r}_O, \qquad \boldsymbol{R} = \boldsymbol{R}_O - \boldsymbol{R}_K, \tag{3}$$

where O is the expansion centre and K is the distant nucleus. The long-distance condition is

$$|\boldsymbol{r}_O| \ll R. \tag{4}$$

Conventions and domain of validity

We use $c = 1/\alpha$ and the scaled vector potentials

$$\boldsymbol{A}^B = \tfrac{1}{2}\boldsymbol{B} \times \boldsymbol{r}_O, \qquad \boldsymbol{A}^K = \alpha^2 \boldsymbol{M}_K \times \frac{\boldsymbol{r}_K}{r_K^3}.$$

The electronic state is assumed to be localized around O, with negligible overlap at the distant nucleus K. Therefore the statement $\mathrm{FC}^{\mathrm{LD}} = 0$ refers to the retained long-distance operator; it does not claim that the exact molecular density at K is identically zero.

What we get: a fixed protocol, used everywhere below without exception:

1. write the field-dependent Hamiltonian;
2. differentiate with respect to B_m or $M_{K,l}$;
3. set the external perturbation to zero;
4. only for the distant nucleus, perform the Taylor expansion around O;
5. apply the RKB/AO representation.

2 Algebraic conventions used in every step

What this step is for

Nothing new is derived in this section. It is the toolbox: three algebraic facts that every later manipulation reduces to, collected once so the appendices can simply point back here instead of re-deriving them each time.

The following identities are used throughout:

$$(\boldsymbol{u} \times \boldsymbol{v})_l = \varepsilon_{lij} u_i v_j, \qquad p_k = -\mathrm{i}\partial_k, \tag{5}$$

$$\sigma_i \sigma_j = \delta_{ij} + \mathrm{i}\varepsilon_{ijk}\sigma_k, \qquad p_k F = F p_k - \mathrm{i}\partial_k F. \tag{6}$$

The singular Coulomb kernel has the distributional derivative

$$\boxed{\partial_j \frac{r_i}{r^3} = \mathrm{PV}\left(\frac{\delta_{ij} r^2 - 3 r_i r_j}{r^5}\right) + \frac{4\pi}{3}\delta_{ij}\delta^{(3)}(\boldsymbol{r}).} \tag{7}$$

Why this one matters: Eq. (7) is the single equation ultimately responsible for the local Fermi-contact term appearing later — every delta function in this paper traces back to it.

Detailed derivation: Appendix A.1.

3 Four-component parent operators

Goal of this step

Write down the two "raw" magnetic-derivative operators we are going to manipulate for the rest of the paper: the response of the Dirac Hamiltonian to a uniform external field $\boldsymbol{B}$, and its response to the nuclear magnetic moment $\boldsymbol{M}_K$.

What we get: two four-component operators that look almost the same in shape, but differ in one crucial way (see below).

In the chosen convention,

$$h^D(\boldsymbol{A}) = \begin{pmatrix} V & c\,\boldsymbol{\sigma}\cdot(\boldsymbol{p}+\boldsymbol{A}) \\ c\,\boldsymbol{\sigma}\cdot(\boldsymbol{p}+\boldsymbol{A}) & V-2c^2 \end{pmatrix}. \tag{8}$$

For the homogeneous magnetic field,

$$\boxed{\mathcal{B}_m \equiv \left(\frac{\partial h^D}{\partial B_m}\right)^{(0)} = \frac{c}{2}\begin{pmatrix} 0 & Q_m^B \\ Q_m^B & 0 \end{pmatrix}, \qquad Q_m^B = (\boldsymbol{r}_O \times \boldsymbol{\sigma})_m.} \tag{9}$$

For the nuclear magnetic moment,

$$\boxed{\mathcal{M}_{K,l} \equiv \left(\frac{\partial h^D}{\partial M_{K,l}}\right)^{(0)} = c\alpha^2\begin{pmatrix} 0 & Q_l^K \\ Q_l^K & 0 \end{pmatrix}, \qquad Q_l^K = \left(\frac{\boldsymbol{r}_K}{r_K^3} \times \boldsymbol{\sigma}\right)_l.} \tag{10}$$

The one difference that drives the whole paper

Q_m^B is built from $\boldsymbol{r}_O$: perfectly smooth everywhere in the electronic region. Q_l^K is built from $\boldsymbol{r}_K/r_K^3$: singular right at the distant nucleus K. Every later result — why FC disappears, why SD survives, why a gauge term appears — is a consequence of this single structural difference.

Detailed derivation: Appendix A.2.

4 RKB transformation: why upper and lower blocks differ

Goal of this step

The Dirac operator is purely off-diagonal (upper–lower). To compute anything in a finite basis we sandwich it with the restricted-kinetic-balance (RKB) transform X^{RKB}. We need the general rule for what that sandwich does to *any* off-diagonal operator, once and for all.

What we get: a simple rule: squeezing an operator of the generic shape $\mathcal{O}[Q,\lambda]$ through X^{RKB} produces a new upper block $U[Q] = Q\,\boldsymbol{\sigma}\cdot\boldsymbol{p}$ and a new lower block $L[Q] = \boldsymbol{\sigma}\cdot\boldsymbol{p}\,Q$ — and these two are *not* the same operator, because in L the momentum operator differentiates the (possibly singular) kernel Q, while in U it does not.

The RKB embedding is

$$X^{\mathrm{RKB}} = \begin{pmatrix} 1 & 0 \\ 0 & \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{2c} \end{pmatrix}. \tag{11}$$

For a generic off-diagonal four-component operator

$$\mathcal{O}[Q,\lambda] = \lambda\begin{pmatrix} 0 & Q \\ Q & 0 \end{pmatrix},$$

the RKB result is

$$\boxed{(X^{\mathrm{RKB}})^\dagger \mathcal{O}[Q,\lambda] X^{\mathrm{RKB}} = \frac{\lambda}{2c}\begin{pmatrix} 0 & Q\,\boldsymbol{\sigma}\cdot\boldsymbol{p} \\ \boldsymbol{\sigma}\cdot\boldsymbol{p}\,Q & 0 \end{pmatrix}.} \tag{12}$$

We therefore define

$$U[Q] = Q\,\boldsymbol{\sigma}\cdot\boldsymbol{p}, \qquad L[Q] = \boldsymbol{\sigma}\cdot\boldsymbol{p}\,Q. \tag{13}$$

The lower block is the one in which momentum differentiates the coordinate-dependent kernel — this is the mechanism that will manufacture the Fermi-contact delta function in Section 5.

Detailed derivation: Appendix A.3.

5 Exact local hyperfine decomposition before any long-distance approximation

Goal of this step

Apply the RKB rule of Section 4 to the *exact*, unapproximated hyperfine kernel $\boldsymbol{a}^K(\boldsymbol{r}) = \boldsymbol{r}_K/r_K^3$ — so we know precisely, term by term, what the true operator looks like before we ever touch the long-distance approximation. This is the reference point everything else in the paper is measured against.

What we get: the exact upper and lower RKB blocks, split into the familiar hyperfine sectors PSO, FC, SD, and a relativistic remainder aREL.

Keep

$$\boldsymbol{a}^K(\boldsymbol{r}) = \frac{\boldsymbol{r}_K}{r_K^3} \tag{14}$$

exactly. For a general vector field $\boldsymbol{a}$, the RKB algebra gives the two generic identities

$$U_l[\boldsymbol{a}] = (\boldsymbol{a}\times\boldsymbol{p})_l + \mathrm{i}[\boldsymbol{a}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l, \tag{15}$$

$$L_l[\boldsymbol{a}] = (\boldsymbol{a}\times\boldsymbol{p})_l - \mathrm{i}[\boldsymbol{a}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l + \mathrm{i}(\nabla\times\boldsymbol{a})_l + (\nabla\cdot\boldsymbol{a})\sigma_l - (\partial_l a_m)\sigma_m. \tag{16}$$

Detailed derivation: Appendix A.4.

For the exact singular field $\boldsymbol{a}^K = \boldsymbol{r}_K/r_K^3$,

$$\nabla\times\boldsymbol{a}^K = 0, \qquad \nabla\cdot\boldsymbol{a}^K = 4\pi\delta^{(3)}(\boldsymbol{r}_K), \tag{17}$$

and the derivative sector of the lower block becomes

$$\mathrm{PV}\left(\frac{3r_{K,l}(\boldsymbol{r}_K\cdot\boldsymbol{\sigma}) - r_K^2\sigma_l}{r_K^5}\right) + \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_K)\sigma_l. \tag{18}$$

Detailed derivation: Appendix A.5.

This identifies

$$\frac{1}{2}\mathrm{PSO}_l \equiv \frac{(\boldsymbol{r}_K\times\boldsymbol{p})_l}{r_K^3}, \tag{19}$$

$$a\mathrm{REL}_l \equiv \frac{\mathrm{i}}{r_K^3}\left[(\boldsymbol{r}_K\cdot\boldsymbol{\sigma})p_l - (\boldsymbol{r}_K\cdot\boldsymbol{p})\sigma_l\right], \tag{20}$$

$$\mathrm{FC}_l \equiv \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_K)\sigma_l, \tag{21}$$

$$\mathrm{SD}_l \equiv \mathrm{PV}\left(\frac{3r_{K,l}(\boldsymbol{r}_K\cdot\boldsymbol{\sigma}) - r_K^2\sigma_l}{r_K^5}\right). \tag{22}$$

What we get: the exact local RKB blocks

$$\boxed{U_l^K = \frac{1}{2}\mathrm{PSO}_l + a\mathrm{REL}_l, \qquad L_l^K = \frac{1}{2}\mathrm{PSO}_l + \mathrm{FC}_l + \mathrm{SD}_l - a\mathrm{REL}_l.} \tag{23}$$

Here PSO, FC, and SD denote the orbital, Fermi-contact, and spin-dipole HFC sectors. OZ and SZ denote the orbital- and spin-Zeeman sectors, with $\mathrm{SZ}_m \equiv \sigma_m$ in the present notation. The symbols aREL and gREL are the corresponding RKB relativistic remainders.

Exact origin of FC

The contact term is created by the delta distribution in $\partial_l(r_{K,m}/r_K^3)$ together with the exact divergence $\nabla\cdot(\boldsymbol{r}_K/r_K^3) = 4\pi\delta^{(3)}(\boldsymbol{r}_K)$. It appears in the lower RKB block because only $\boldsymbol{\sigma}\cdot\boldsymbol{p}Q_l^K$ differentiates the singular kernel.

The complete exact local AO/RKB result is therefore

$$\mathcal{A}_l^{K,\mathrm{local}} = \frac{\alpha^2}{2} D^0_{a\kappa,b\eta}(Jv) \int \chi_\eta^* \begin{pmatrix} 0 & \frac{1}{2}\mathrm{PSO}_l + a\mathrm{REL}_l \\ \frac{1}{2}\mathrm{PSO}_l + \mathrm{FC}_l + \mathrm{SD}_l - a\mathrm{REL}_l & 0 \end{pmatrix}_{ba} \chi_\kappa \,\mathrm{d}V. \tag{24}$$

6 Long-distance expansion and the exact step at which FC disappears

Goal of this step

Replace the exact, singular field $\boldsymbol{r}_K/r_K^3$ by its Taylor expansion around O, valid when the electron is far from nucleus K, and track — term by term — which piece of the exact result (Eq. (23)) survives this replacement and which one is destroyed.

What we get: the smooth long-distance field $\boldsymbol{a}^{\mathrm{LD}}$, and the single most important qualitative fact in the paper: because this field has no delta-function source, *the Fermi-contact term is gone*, while the spin-dipole term survives and turns into $T\cdot\mathrm{SZ}$.

Write

$$\boldsymbol{a}^K(\boldsymbol{r}) = \frac{\boldsymbol{R}+\boldsymbol{r}_O}{|\boldsymbol{R}+\boldsymbol{r}_O|^3}. \tag{25}$$

To first order in r_O/R,

$$\boxed{\boldsymbol{a}^{\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} + \frac{\boldsymbol{r}_O}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{R}}{R^5} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_O.} \tag{26}$$

Detailed derivation: Appendix A.6.

Because $\boldsymbol{R}$ and $T_{ml}(\boldsymbol{R})$ are constants with respect to $\boldsymbol{r}_O$,

$$\boxed{\partial_l a_m^{\mathrm{LD}} = -T_{ml}, \qquad \nabla\times\boldsymbol{a}^{\mathrm{LD}} = 0, \qquad \nabla\cdot\boldsymbol{a}^{\mathrm{LD}} = 0.} \tag{27}$$

Substitution into Eq. (16) gives the surviving derivative term

$$-(\partial_l a_m^{\mathrm{LD}})\sigma_m = \sum_m T_{ml}\sigma_m. \tag{28}$$

The decisive approximation

The FC term disappears at the replacement

$$\frac{\boldsymbol{r}_K}{r_K^3} \quad\longrightarrow\quad \boldsymbol{a}^{\mathrm{LD}}(\boldsymbol{r}_O) = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_O.$$

The exact field is singular at K; the retained Taylor field is smooth in the electronic region around O. Therefore its derivatives contain no delta distribution. There is no later cancellation of FC — it is simply absent from the moment the field is replaced.

What we get

$$\boxed{\mathrm{FC}_l^{\mathrm{LD}} = 0, \qquad \mathrm{SD}_l^{\mathrm{LD}} = \sum_m T_{ml}\,\mathrm{SZ}_m.} \tag{29}$$

7 Direct comparison: exact local field versus long-distance field

Purpose of this table

Put the exact calculation (Section 5) and the long-distance calculation (Section 6) side by side, so the moment where FC disappears is visible at a glance, instead of buried in algebra.

Both calculations use the same generic lower-block identity, Eq. (16); only the inserted vector field changes:

$$\frac{\boldsymbol{r}_K}{r_K^3} \quad\longrightarrow\quad \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}\boldsymbol{r}_O.$$

	Exact local kernel $\boldsymbol{r}_K/r_K^3$	**Long-distance Taylor field**
Curl	0	0
Divergence	$4\pi\delta^{(3)}(\boldsymbol{r}_K)$	0
Gradient	regular SD tensor + delta part	constant $-T_{ml}(\boldsymbol{R})$
Lower spin sector	$\mathrm{FC}_l + \mathrm{SD}_l$	$T_{ml}\sigma_m$
Contact term	$\frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_K)\sigma_l$	none

8 Explicit RKB blocks after the long-distance expansion

Goal of this step

Assemble the complete upper and lower RKB blocks using the long-distance field $\boldsymbol{a}^{\mathrm{LD}}$ — the two full expressions we are about to split into “Zeeman” and “gauge” pieces in the next section.

What we get: explicit, fully written-out U_l^{LD} and L_l^{LD}, with no delta functions anywhere — confirming, in a different form, the $\mathrm{FC}^{\mathrm{LD}} = 0$ result just found.

The long-distance two-component kernel is

$$Q_l^{K,\mathrm{LD}} = \frac{(\boldsymbol{R}\times\boldsymbol{\sigma})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{\sigma})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{\sigma})_l}{R^5}. \tag{30}$$

The upper block is

$$\begin{aligned} U_l^{\mathrm{LD}} = {}& \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} \\ &+ \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5}. \end{aligned} \tag{31}$$

The lower block is

$$\begin{aligned} L_l^{\mathrm{LD}} = {}& \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} \\ &- \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} + \sum_m T_{ml}\sigma_m. \end{aligned} \tag{32}$$

These are the complete retained long-distance blocks; in particular no $\delta^{(3)}(\boldsymbol{r}_K)$ occurs.

Detailed derivation: Appendix A.7.

9 Decomposition into a transferred Zeeman operator and a gauge term

Goal of this whole section

This is the central section of the paper. **We want to show** that the long-distance hyperfine kernel found in Section 8 is *nothing but* the ordinary Zeeman kernel, redirected by the geometric tensor $T(\boldsymbol{R})$, plus a leftover piece — and that this leftover piece is a pure gauge term that must vanish once we evaluate it in a real (stationary) electronic state. If this holds, the entire distant-nucleus hyperfine response collapses onto quantities (the g-tensor) we already know how to compute.

9.1 Zeeman blocks

Goal of this step

Before comparing to the hyperfine kernel, get the RKB blocks of the plain Zeeman operator itself, using the same machinery as before.

What we get: U_m^B and L_m^B, split into orbital Zeeman (OZ), spin Zeeman (SZ), and a relativistic remainder gREL — these are the building blocks the hyperfine kernel will be measured against.

For

$$Q_m^B = (\boldsymbol{r}_O \times \boldsymbol{\sigma})_m,$$

the RKB blocks are

$$U_m^B = (\boldsymbol{r}_O \times \boldsymbol{p})_m + \mathrm{i}[\boldsymbol{r}_O \times (\boldsymbol{p} \times \boldsymbol{\sigma})]_m, \tag{33}$$

$$L_m^B = (\boldsymbol{r}_O \times \boldsymbol{p})_m - \mathrm{i}[\boldsymbol{r}_O \times (\boldsymbol{p} \times \boldsymbol{\sigma})]_m + 2\sigma_m. \tag{34}$$

Thus

$$\boxed{U_m^B = \mathrm{OZ}_m + g\mathrm{REL}_m, \qquad L_m^B = \mathrm{OZ}_m + 2\mathrm{SZ}_m - g\mathrm{REL}_m.} \tag{35}$$

Detailed derivation: Appendix A.8.

9.2 Kernel-level decomposition

Goal of this step

Show, at the level of the bare kernel $Q_l^{K,\mathrm{LD}}$ (before worrying about p, $\boldsymbol{\sigma} \cdot \boldsymbol{p}$, or any of the RKB machinery), that it splits exactly into a T-rescaled copy of the Zeeman kernel Q_m^B plus a remainder Q_l^G.

What we get: an *exact* algebraic identity — verified independently, term by term — with no approximation beyond the long-distance Taylor expansion already made in Section 6.

The long-distance hyperfine kernel can be written as

$$\boxed{Q_l^{K,\mathrm{LD}} = \tfrac{1}{2} \sum_m T_{ml} Q_m^B + \frac{1}{\alpha^2} Q_l^G,} \tag{36}$$

where

$$Q_l^G = \alpha^2 \left[\frac{(\boldsymbol{R} \times \boldsymbol{\sigma})_l}{R^3} - \frac{3}{2R^5} \Big((\boldsymbol{\sigma} \cdot \boldsymbol{R})(\boldsymbol{R} \times \boldsymbol{r}_O)_l + (\boldsymbol{R} \cdot \boldsymbol{r}_O)(\boldsymbol{R} \times \boldsymbol{\sigma})_l \Big) \right]. \tag{37}$$

9.3 Gauge remainder as a commutator

Goal of this step

Show that the leftover piece Q_l^G is not some arbitrary residual term, but is exactly a *gradient* $\boldsymbol{\sigma} \cdot \nabla f_l$ of a scalar function f_l — which is precisely the algebraic signature of a gauge transformation, and is what lets us later prove it vanishes for physical states.

What we get: an explicit gauge-generating function $f_l(\boldsymbol{r}_O)$, and the identity $Q_l^G = \boldsymbol{\sigma} \cdot \nabla f_l$, which promotes the whole remainder to a commutator of $h^{D,(0)}$ with f_l.

Define

$$\boxed{f_l(\boldsymbol{r}_O) = \alpha^2 \left[\frac{(\boldsymbol{R} \times \boldsymbol{r}_O)_l}{R^3} - \frac{3}{2R^5} (\boldsymbol{R} \cdot \boldsymbol{r}_O)(\boldsymbol{R} \times \boldsymbol{r}_O)_l \right].} \tag{38}$$

Then

$$Q_l^G = \boldsymbol{\sigma} \cdot \nabla f_l, \tag{39}$$

and the four-component gauge remainder is

$$\boxed{\mathrm{i}[h^{D,(0)}, f_l] = c \begin{pmatrix} 0 & Q_l^G \\ Q_l^G & 0 \end{pmatrix}.} \tag{40}$$

Detailed derivation: Appendix A.9.

The complete RKB representation of the commutator is

$$\boxed{(X^{\mathrm{RKB}})^\dagger \, \mathrm{i}[h^{D,(0)}, f_l] X^{\mathrm{RKB}} = \frac{1}{2} \begin{pmatrix} 0 & U_l^G \\ L_l^G & 0 \end{pmatrix}.} \tag{41}$$

The individual gauge blocks are nonzero; they must be treated together as one commutator.

9.4 Stationary-state cancellation

Goal of this step

Prove that whatever Q_l^G (equivalently f_l) actually looks like in detail, its contribution to any real, physically measurable matrix element is exactly zero — so it can never show up in an observable, however messy it looks on paper.

What we get: a clean vanishing theorem, proved three independent ways (eigenstate argument, density-matrix trace, and current continuity) so the result does not depend on which picture you find most convincing.

For a stationary eigenstate,

$$\boxed{\langle \phi_i^0 | \mathrm{i}[h^{D,(0)}, f_l] | \phi_i^0 \rangle = 0.} \tag{42}$$

The same statement can be written as

$$\mathrm{Tr}[D^0 \, \mathrm{i}[h^{D,(0)}, f_l]] = 0$$

or as the current integral

$$- \int \boldsymbol{j}^0 \cdot \nabla f_l \, \mathrm{d}V = 0.$$

Detailed derivation: Appendix A.10.

10 From spinors to the AO density contraction

Goal of this step

Pure bookkeeping step: rewrite the spinor matrix element of any RKB-transformed operator in terms of the LCAO expansion coefficients C^0, so the formula is in the form an actual quantum-chemistry calculation uses.

What we get: the standard AO-density-matrix contraction formula — nothing physical changes here, only the notation.

Let

$$\phi^0_{m,i}(J_v) = X^{\mathrm{RKB}}_{m,a\kappa} C^0_{a\kappa,i}(J_v), \tag{43}$$

and define

$$D^0_{a\kappa,b\eta}(J_v) = C^0_{a\kappa,i}(J_v)[C^0_{b\eta,i}(J_v)]^*. \tag{44}$$

Then for an RKB-transformed one-electron operator,

$$\boxed{\langle \phi^0_i | \widetilde{\mathcal{O}} | \phi^0_i \rangle = D^0_{a\kappa,b\eta}(J_v) \int \chi^*_\eta \widetilde{\mathcal{O}}_{ba} \chi_\kappa \, \mathrm{d}V.} \tag{45}$$

Detailed derivation: Appendix A.11.

11 Final RKB/AO identity

Goal of this step

Combine the kernel-level split of Section 9 with the AO contraction of Section 10 into one identity, written directly in the blocks U, L that a real calculation manipulates.

What we get: the operator identity below — and, because $U[\,\cdot\,]$ and $L[\,\cdot\,]$ are linear in the kernel Q, this follows automatically from Eq. (36): no new algebra is needed, it is the same split, just carried through to U and L.

From Eq. (36),

$$U^{\mathrm{LD}}_l = \tfrac{1}{2} T_{ml} U^B_m + \alpha^{-2} U^G_l, \qquad L^{\mathrm{LD}}_l = \tfrac{1}{2} T_{ml} L^B_m + \alpha^{-2} L^G_l. \tag{46}$$

Including the different parent prefactors,

$$\boxed{\frac{\alpha^2}{2} \begin{pmatrix} 0 & U^{\mathrm{LD}}_l \\ L^{\mathrm{LD}}_l & 0 \end{pmatrix} = \alpha^2 T_{ml} \frac{1}{4} \begin{pmatrix} 0 & U^B_m \\ L^B_m & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & U^G_l \\ L^G_l & 0 \end{pmatrix}.} \tag{47}$$

At the AO/density-contracted level,

$$\begin{aligned} &\frac{\alpha^2}{2} D^0_{a\kappa,b\eta} \int \chi^*_\eta \begin{pmatrix} 0 & U^{\mathrm{LD}}_l \\ L^{\mathrm{LD}}_l & 0 \end{pmatrix}_{ba} \chi_\kappa \, \mathrm{d}V \\ &\quad = \alpha^2 T_{ml} \frac{1}{4} D^0_{a\kappa,b\eta} \int \chi^*_\eta \begin{pmatrix} 0 & U^B_m \\ L^B_m & 0 \end{pmatrix}_{ba} \chi_\kappa \, \mathrm{d}V + \frac{1}{2} D^0_{a\kappa,b\eta} \int \chi^*_\eta \begin{pmatrix} 0 & U^G_l \\ L^G_l & 0 \end{pmatrix}_{ba} \chi_\kappa \, \mathrm{d}V. \end{aligned} \tag{48}$$

For a stationary state the final term vanishes only after the complete upper-plus-lower matrix element is formed.

12 Term-by-term interpretation

Purpose of this table

Translate every boxed equation above into a one-line, plain-language statement of what happens to each named hyperfine sector (PSO, FC, SD, aREL) once the long-distance limit is taken.

Local label	Long-distance image	Meaning
PSO	$T \cdot$ OZ + gauge rearrangement	PSO is not separately equal to the transferred OZ piece before the gauge block is included.
FC	0	The delta source disappears when the singular field is replaced by the smooth Taylor field.
SD	$T \cdot$ SZ	The surviving smooth derivative is $T_{ml}\sigma_m$.
aREL	$T \cdot g$REL + gauge rearrangement	The relativistic orbital remainder is gauge-sensitive.

The safest operator statement is therefore

$$\boxed{\text{distant-nucleus HFC operator} = T(\boldsymbol{R}) \cdot \text{complete Zeeman operator} + \text{gauge commutator.}} \tag{49}$$

13 Compact derivation chain

Purpose of this section

Everything above, compressed into five lines, for a reader (or your future self) who only wants the skeleton of the argument.

The whole RKB argument is

$$\frac{\boldsymbol{r}_K}{r_K^3} \xrightarrow{|\boldsymbol{r}_O| \ll R} \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_O, \tag{50}$$

$$\partial_l a_m^{\mathrm{LD}} = -T_{ml}, \qquad \nabla \cdot \boldsymbol{a}^{\mathrm{LD}} = 0, \qquad \nabla \times \boldsymbol{a}^{\mathrm{LD}} = 0, \tag{51}$$

$$\mathrm{FC}^{\mathrm{LD}} = 0, \qquad \mathrm{SD}^{\mathrm{LD}} = T \cdot \mathrm{SZ}, \tag{52}$$

$$Q_l^{K,\mathrm{LD}} = \tfrac{1}{2} T_{ml} Q_m^B + \alpha^{-2} Q_l^G, \qquad Q_l^G = \boldsymbol{\sigma} \cdot \nabla f_l, \tag{53}$$

$$\left(\frac{\partial h^D}{\partial M_{K,l}}\right)^{(0)}_{\mathrm{LD}} \simeq \alpha^2 T_{ml} \left(\frac{\partial h^D}{\partial B_m}\right)^{(0)} + \mathrm{i}[h^{D,(0)}, f_l]. \tag{54}$$

Complete AO-sandwiched RKB map: exactly which long-distance terms become Zeeman terms and which terms belong to the gauge block

Purpose of this map

Nothing new is derived here. This is the same U_l^{LD} and L_l^{LD} from Section 8, expanded so that literally *every single term* is labelled: which underlying local sector it came from (PSO, aREL, FC, SD), and which piece of the final split (Zeeman or gauge, Eq. (36)) it belongs to. Use this page as the term-by-term audit trail if you ever need to check a specific piece.

Define

$$\boxed{\mathcal{C}_v[X] \equiv D^0_{a\kappa,b\eta}(J_v) \int \chi_\eta^* \, [X]_{ba} \, \chi_\kappa \, \mathrm{d}V.} \tag{55}$$

Upper RKB block: $U_l^{\mathrm{LD}} = \frac{1}{2} T_{ml} U_m^B + \alpha^{-2} U_l^G$

$$
\begin{aligned}
\mathcal{U}_l^{\mathrm{LD}}(J_v) \equiv \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{\frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5}}_{\frac{1}{2}\mathrm{PSO}_l^{\mathrm{LD}}} \\
& + \underbrace{\frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5}}_{a\mathrm{REL}_l^{\mathrm{LD}}}\Bigg] \\
= \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{-\frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} + \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p})}_{\frac{1}{2}T_{ml}\mathrm{OZ}_m} + \underbrace{-\frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} + \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]}_{\frac{1}{2}T_{ml}g\mathrm{REL}_m}\Bigg] \\
+ \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{\frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{3(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p})}_{\alpha^{-2}U^G_{l,\mathrm{orb}}} \\
& + \underbrace{\frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{3\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} - \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]}_{\alpha^{-2}U^G_{l,\mathrm{rel}}}\Bigg].
\end{aligned}
\tag{56}
$$

Lower RKB block: the same rearrangement plus the direct $\mathrm{SD} \to T\cdot\mathrm{SZ}$ map

$$
\begin{aligned}
\mathcal{L}_l^{\mathrm{LD}}(J_v) \equiv \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{\frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5}}_{\frac{1}{2}\mathrm{PSO}_l^{\mathrm{LD}}} + \underbrace{0}_{\mathrm{FC}_l^{\mathrm{LD}}} + \underbrace{\sum_m T_{ml}\sigma_m}_{\mathrm{SD}_l^{\mathrm{LD}}} \\
& - \underbrace{\left[\frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5}\right]}_{a\mathrm{REL}_l^{\mathrm{LD}}}\Bigg] \\
= \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{-\frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} + \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p})}_{\frac{1}{2}T_{ml}\mathrm{OZ}_m} + \underbrace{\sum_m T_{ml}\sigma_m}_{\frac{1}{2}T_{ml}(2\mathrm{SZ}_m)} \\
& + \underbrace{\frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} - \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]}_{-\frac{1}{2}T_{ml}g\mathrm{REL}_m}\Bigg] \\
+ \frac{\alpha^2}{2}\,\mathcal{C}_v\Bigg[& \underbrace{\frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{3(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p})}_{\alpha^{-2}L^G_{l,\mathrm{orb}}} \\
& + \underbrace{-\frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - \frac{3\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} + 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} + \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]}_{\alpha^{-2}L^G_{l,\mathrm{rel}}}\Bigg]
\end{aligned}
\tag{57}
$$

Complete physical matrix element

$$\boxed{\begin{aligned}&\frac{\alpha^2}{2}\mathcal{C}_v\left[\begin{pmatrix}0 & U_l^{\mathrm{LD}}\\ L_l^{\mathrm{LD}} & 0\end{pmatrix}\right]\\ &\quad=\alpha^2 T_{ml}(\boldsymbol{R})\frac{1}{4}\mathcal{C}_v\left[\begin{pmatrix}0 & \mathrm{OZ}_m+g\mathrm{REL}_m\\ \mathrm{OZ}_m+2\mathrm{SZ}_m-g\mathrm{REL}_m & 0\end{pmatrix}\right]+\underbrace{\frac{1}{2}\mathcal{C}_v\left[\begin{pmatrix}0 & U_l^G\\ L_l^G & 0\end{pmatrix}\right]}_{\langle \mathrm{i}[h^{D,(0)},f_l]\rangle_{J_v}}\\ &\quad=\boxed{\alpha^2 T_{ml}(\boldsymbol{R})\left\langle\left(\frac{\partial h^D}{\partial B_m}\right)^{(0)}\right\rangle_{J_v}}\qquad \text{for a stationary state.}\end{aligned}}\tag{58}$$

Exact reading of the RKB map

$$\boxed{\begin{aligned}\frac{1}{2}\mathrm{PSO}^{\mathrm{LD}}+a\mathrm{REL}^{\mathrm{LD}}&=\frac{1}{2}T\cdot(\mathrm{OZ}+g\mathrm{REL})+G_U,\\ \frac{1}{2}\mathrm{PSO}^{\mathrm{LD}}+\underbrace{\mathrm{FC}^{\mathrm{LD}}}_{0}+\mathrm{SD}^{\mathrm{LD}}-a\mathrm{REL}^{\mathrm{LD}}&=\frac{1}{2}T\cdot(\mathrm{OZ}+2\mathrm{SZ}-g\mathrm{REL})+G_L.\end{aligned}}$$

The spin mapping is direct. The orbital and relativistic pieces require the gauge rearrangement, and G_U+G_L must be recombined into the complete commutator before taking the physical matrix element.

14 Consistency audit

Purpose of this section

A checklist of the levels at which the derivation was cross-checked, so a reader can see at a glance what has and has not been verified.

The derivation is checked at the levels that matter for the final result:

1. Pauli products and Levi–Civita contractions;
2. the distributional derivative and the $8\pi/3$ FC coefficient;
3. the first-order Taylor expansion;
4. the kernel identity $Q_l^{K,\mathrm{LD}}=\frac{1}{2}T_{ml}Q_m^B+\alpha^{-2}Q_l^G$;
5. the distinct RKB prefactors inherited from the hyperfine, Zeeman, and gauge parent operators.

15 Gordon decomposition

Why we redo the whole thing a second way

The RKB derivation above is one specific technical route. **We want** an independent cross-check using a genuinely different method — rewriting the same magnetic-derivative matrix elements in terms of the four-component current instead of two-component blocks — so that if both routes give the same final answer, we can trust the result was not an artefact of the RKB bookkeeping.

15.1 Purpose of the second derivation

The Gordon route deserves the same logical status as the RKB route. It starts from the same field-dependent Hamiltonian and the same two magnetic derivatives, but rewrites their matrix elements in terms of the four-component current. The logical chain is

$$h^D(\boldsymbol{B}, \boldsymbol{\mu}^M) \longrightarrow \left\{ \frac{\partial h^D}{\partial B_u}, \frac{\partial h^D}{\partial \mu_u^M} \right\} \longrightarrow \left\langle \frac{\partial h^D}{\partial \lambda} \right\rangle = -\frac{1}{c} \int \boldsymbol{j} \cdot \frac{\partial \boldsymbol{A}}{\partial \lambda} \, \mathrm{d}V \longrightarrow \boldsymbol{j}_{\mathrm{orb}} + \boldsymbol{j}_{\mathrm{spin}}. \tag{59}$$

15.2 The two derivatives of the Dirac–Kohn–Sham Hamiltonian

Goal of this step

The same first step as Section 3, but now in the 4-component Dirac-matrix $(\boldsymbol{\alpha}, \beta)$

What we get: $\partial h^D / \partial B_u$ and $\partial h^D / \partial \mu_u^M$ in closed form.

$$h^D(\lambda) = c\boldsymbol{\alpha} \cdot \boldsymbol{p} + \boldsymbol{\alpha} \cdot \boldsymbol{A}(\lambda) + c^2 \beta + V_{4\times 4}. \tag{60}$$

With

$$\boldsymbol{A}^B = \frac{1}{2} \boldsymbol{B} \times \boldsymbol{r}_G, \qquad \boldsymbol{A}^M = \boldsymbol{\mu}^M \times \frac{\boldsymbol{r}_M}{r_M^3},$$

the derivatives are

$$\boxed{\frac{\partial h^D}{\partial B_u} = \frac{1}{2} (\boldsymbol{r}_G \times \boldsymbol{\alpha})_u, \qquad \frac{\partial h^D}{\partial \mu_u^M} = \left(\frac{\boldsymbol{r}_M}{r_M^3} \times \boldsymbol{\alpha} \right)_u.} \tag{61}$$

Detailed derivation: Appendix B.1.

For a doublet,

$$g_{uv} = 4c \frac{\mathrm{d}E(J_v)}{\mathrm{d}B_u}, \qquad A_{uv}^M = \frac{\gamma^M}{\pi} \frac{\mathrm{d}E(J_v)}{\mathrm{d}\mu_u^M}. \tag{62}$$

15.3 From Hamiltonian derivatives to the current

Goal of this step

Turn the abstract Hamiltonian derivatives above into something computable: an integral of the physical current density against the derivative of the vector potential.

What we get: the Hellmann–Feynman bridge $\langle \partial h^D / \partial \lambda \rangle = -\frac{1}{c} \int \boldsymbol{j}^0 \cdot \partial \boldsymbol{A} / \partial \lambda \, \mathrm{d}V$, which turns g_{uv} and A_{uv}^M into current integrals.

The field-free Dirac current is

$$\boxed{\boldsymbol{j}^0(J_v; \boldsymbol{r}) = -c \, \phi_i^{0\dagger}(J_v; \boldsymbol{r}) \boldsymbol{\alpha} \phi_i^0(J_v; \boldsymbol{r}).} \tag{63}$$

Hellmann–Feynman gives

$$\boxed{\left\langle \frac{\partial h^D}{\partial \lambda} \right\rangle = -\frac{1}{c} \int \boldsymbol{j}^0 \cdot \frac{\partial \boldsymbol{A}}{\partial \lambda} \, \mathrm{d}V.} \tag{64}$$

Hence

$$g_{uv} = -2\int [\boldsymbol{r}_G \times \boldsymbol{j}^0(J_v)]_u \,\mathrm{d}V, \tag{65}$$

$$A^M_{uv} = -\frac{\gamma^M}{\pi c}\int \left[\frac{\boldsymbol{r}_M}{r_M^3} \times \boldsymbol{j}^0(J_v)\right]_u \mathrm{d}V. \tag{66}$$

15.4 Field-free Gordon decomposition

Goal of this step

Split the exact current $\boldsymbol{j}^0$ itself into an orbital piece and a spin piece — this is the current-picture analogue of splitting the RKB kernel into PSO/FC/SD.

What we get: the Gordon current identity below, with the spin current written as (minus one half) the curl of the spin density.

For a local real scalar potential, the exact field-free Gordon current is

$$\boxed{\boldsymbol{j}^0 = \boldsymbol{j}^0_{\text{orb}} + \boldsymbol{j}^0_{\text{spin}} = -\operatorname{Im}\{\phi_i^{0\dagger}\beta\nabla\phi_i^0\} - \frac{1}{2}\nabla \times \boldsymbol{\rho}^{G,0},} \tag{67}$$

The Gordon decomposition does not introduce separate aREL or gREL labels: those are specific to the RKB block organization. The same relativistic information is already contained in the four-component spinor, β, and the Gordon densities. where

$$\rho_m^{G,0} = \phi_i^{0\dagger}\beta\Sigma_m\phi_i^0. \tag{68}$$

This is an exact reorganization of the same large–small coupling present in $-c\phi^\dagger\boldsymbol{\alpha}\phi$.

Detailed derivation: Appendix B.2.

15.5 The two integration-by-parts identities that generate all tensor terms

Goal of this step

Isolate the only two integrations by parts needed anywhere in the Gordon route — one for the orbital current, one for the spin current — so every later formula (g-tensor, HFC tensor) is a direct application of one of these two, not a fresh calculation.

What we get: Eqs. (69) and (70): the complete IBP toolkit for this section.

For a curl-free geometric field $\boldsymbol{a}$, the orbital-current manipulation gives

$$\boxed{\int \varepsilon_{uwk}a_w\left[\phi^\dagger p_k\beta\phi - (p_k\phi^\dagger)\beta\phi\right]\mathrm{d}V = 2\int \phi^\dagger(\boldsymbol{a}\times\boldsymbol{p})_u\beta\phi\,\mathrm{d}V.} \tag{69}$$

For the spin current,

$$\boxed{\int [\boldsymbol{a}\times(\nabla\times\boldsymbol{\rho})]_u\,\mathrm{d}V = \int [(\partial_m a_m)\rho_u - (\partial_u a_n)\rho_n]\,\mathrm{d}V.} \tag{70}$$

These two identities are the only integrations by parts required to generate OZ/SZ and PSO/FC/SD.

Detailed derivation: Appendix B.3.

15.6 Gordon decomposition of the g-tensor

Goal of this step

Apply the orbital identity to $\boldsymbol{a} = \boldsymbol{r}_G$ and the spin identity to get the g-tensor in closed form — the Gordon-route counterpart of Eq. (35).

What we get: $g_{uv} = g_{uv}^{\mathrm{OZ}} + g_{uv}^{\mathrm{SZ}}$, a clean orbital/spin split with no leftover term.

Equation (69) with $\boldsymbol{a} = \boldsymbol{r}_G$ gives

$$\boxed{g_{uv}^{\mathrm{OZ}} = 2 \int \phi_i^{0\dagger}(J_v)(\boldsymbol{r}_G \times \boldsymbol{p})_u \beta \phi_i^0(J_v)\,\mathrm{d}V.} \tag{71}$$

Equation (70) gives

$$\boxed{g_{uv}^{\mathrm{SZ}} = 2 \int \rho_u^{G,0}(J_v)\,\mathrm{d}V.} \tag{72}$$

Therefore

$$\boxed{g_{uv} = g_{uv}^{\mathrm{OZ}} + g_{uv}^{\mathrm{SZ}}.} \tag{73}$$

15.7 Exact local Gordon decomposition of the A-tensor

Goal of this step

Same idea as Section 5, but now for the current picture: get the *exact* hyperfine A-tensor, split into PSO/FC/SD, before any long-distance approximation.

What we get: the exact Gordon-route PSO, FC, and SD terms — and, reassuringly, the same $8\pi/3$ FC coefficient found earlier by the completely different RKB route.

For the exact nuclear field

$$\boldsymbol{a}^M = \frac{\boldsymbol{r}_M}{r_M^3},$$

the orbital current gives

$$\boxed{A_{uv}^{M,\mathrm{PSO}} = \frac{\gamma^M}{2\pi c} \int \phi_i^{0\dagger}(J_v) 2 \left[\frac{\boldsymbol{r}_M \times \boldsymbol{p}}{r_M^3}\right]_u \beta \phi_i^0(J_v)\,\mathrm{d}V.} \tag{74}$$

Before splitting the spin contribution, one has

$$A_{uv}^{M,\mathrm{spin}} = \frac{\gamma^M}{2\pi c} \int \left[(\partial_w a_w^M)\rho_u^{G,0} - (\partial_u a_w^M)\rho_w^{G,0}\right]\mathrm{d}V. \tag{75}$$

Using the exact distributional derivatives gives

$$A_{uv}^{M,\mathrm{FC}} = \boxed{\frac{\gamma^M}{2\pi c} \int \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0}\,\mathrm{d}V,} \tag{76}$$

$$A_{uv}^{M,\mathrm{SD}} = \boxed{\frac{\gamma^M}{2\pi c} \int \frac{3(r_M)_u(r_M)_w - \delta_{uw} r_M^2}{r_M^5}\rho_w^{G,0}\,\mathrm{d}V.} \tag{77}$$

Thus

$$\boxed{A_{uv}^M = A_{uv}^{M,\mathrm{PSO}} + A_{uv}^{M,\mathrm{FC}} + A_{uv}^{M,\mathrm{SD}}.} \tag{78}$$

Detailed derivation: Appendix B.4.

15.8 Long-distance reduction of the Gordon terms

Goal of this step

Repeat the long-distance replacement of Section 6, now on the Gordon-route A-tensor, and identify the current-picture version of the gauge term f_l.

What we get: $A_{uv}^{M,\mathrm{FC,LD}} = 0$ again, a clean PSO/SD mapping onto $g^{\mathrm{OZ}}/g^{\mathrm{SZ}}$, and an explicit proof that the gauge remainder integrates to zero for a stationary state — the current-picture mirror of Section 9.

Write

$$\boldsymbol{r}_M = \boldsymbol{R} + \boldsymbol{r}_G, \qquad \boldsymbol{a}^{M,\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_G. \tag{79}$$

Then

$$\partial_u a_w^{M,\mathrm{LD}} = -T_{wu}, \qquad \partial_w a_w^{M,\mathrm{LD}} = 0, \qquad \nabla \times \boldsymbol{a}^{M,\mathrm{LD}} = 0. \tag{80}$$

The vector-potential derivative can be separated into a uniform-field part and a gradient:

$$\boxed{\boldsymbol{e}_u \times \boldsymbol{a}^{M,\mathrm{LD}} = \frac{1}{2}T_{ku}(\boldsymbol{R})(\boldsymbol{e}_k \times \boldsymbol{r}_G) + \nabla\chi_u,} \tag{81}$$

where

$$\chi_u = \frac{(\boldsymbol{R} \times \boldsymbol{r}_G)_u}{R^3} - \frac{3}{2R^5}(\boldsymbol{R}\cdot\boldsymbol{r}_G)(\boldsymbol{R}\times\boldsymbol{r}_G)_u. \tag{82}$$

Detailed derivation: Appendix B.5.

For a localized stationary state,

$$\boxed{\int \boldsymbol{j}^0 \cdot \nabla\chi_u \,\mathrm{d}V = 0.} \tag{83}$$

For the same expansion centre and gauge origin, $\boldsymbol{r}_O = \boldsymbol{r}_G$,

$$\boxed{f_u = \alpha^2\chi_u.} \tag{84}$$

Thus the RKB commutator and the Gordon gradient integral are two representations of the same gauge remainder.

Detailed derivation: Appendix B.6.

The orbital part therefore maps as

$$\boxed{A_{uv}^{M,\mathrm{PSO,LD}} = \frac{\gamma^M}{4\pi c}T_{ku}(\boldsymbol{R})g_{kv}^{\mathrm{OZ}}.} \tag{85}$$

The spin sector must be reduced from the unsplit expression Eq. (75). Since the long-distance field is smooth,

$$\boxed{A_{uv}^{M,\mathrm{FC,LD}} = 0,} \tag{86}$$

while

$$\boxed{A_{uv}^{M,\mathrm{SD,LD}} = \frac{\gamma^M}{4\pi c}T_{wu}(\boldsymbol{R})g_{wv}^{\mathrm{SZ}}.} \tag{87}$$

Thus the Gordon term-by-term map is

$$\mathrm{PSO} \longrightarrow \frac{\gamma^M}{4\pi c}\boldsymbol{T}\cdot\boldsymbol{g}^{\mathrm{OZ}}, \tag{88}$$

$$\mathrm{FC} \longrightarrow 0, \tag{89}$$

$$\mathrm{SD} \longrightarrow \frac{\gamma^M}{4\pi c}\boldsymbol{T}\cdot\boldsymbol{g}^{\mathrm{SZ}}, \tag{90}$$

and

$$\boxed{A_{uv}^{M,\mathrm{LD}} = \frac{\gamma^M}{4\pi c}T_{ku}(\boldsymbol{R})g_{kv}.} \tag{91}$$

Complete one-equation map of all Gordon terms in the long-distance limit

Purpose of this map

Same idea as the RKB giant map on p. 12: nothing new, just every term of the Gordon-route A-tensor labelled explicitly, so the PSO→ T·OZ, FC→ 0, SD→ T·SZ statements above can be checked term by term rather than taken on faith.

$$\begin{aligned}
A^M_{uv} &= \underbrace{\frac{\gamma^M}{2\pi c}\int \phi_i^{0\dagger}(J_v) 2\left[\frac{\boldsymbol{r}_M\times\boldsymbol{p}}{r_M^3}\right]_u \beta\phi_i^0(J_v)\,\mathrm{d}V}_{\mathrm{PSO}\leftarrow \boldsymbol{j}^0_{\mathrm{orb}}} \\
&\quad + \underbrace{\frac{\gamma^M}{2\pi c}\int \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0}(J_v)\,\mathrm{d}V}_{\mathrm{FC}} + \underbrace{\frac{\gamma^M}{2\pi c}\int \frac{3(r_M)_u(r_M)_w-\delta_{uw}r_M^2}{r_M^5}\rho_w^{G,0}(J_v)\,\mathrm{d}V}_{\mathrm{SD}} \\
&\qquad \xrightarrow[\partial_u a_w^{M,\mathrm{LD}}=-T_{wu},\ \partial_w a_w^{M,\mathrm{LD}}=0]{\boldsymbol{a}^M\mapsto\boldsymbol{a}^{M,\mathrm{LD}},\quad \boldsymbol{e}_u\times\boldsymbol{a}^{M,\mathrm{LD}}=\frac{1}{2}T_{ku}(\boldsymbol{e}_k\times\boldsymbol{r}_G)+\nabla\chi_u} \\
A^{M,\mathrm{LD}}_{uv} &= \underbrace{-\frac{\gamma^M}{\pi c}\int \boldsymbol{j}^0_{\mathrm{orb}}\cdot\left[\frac{1}{2}T_{ku}(\boldsymbol{e}_k\times\boldsymbol{r}_G)+\nabla\chi_u\right]\mathrm{d}V}_{\mathrm{PSO^{LD}}} + \underbrace{0}_{\mathrm{FC^{LD}}} + \underbrace{\frac{\gamma^M}{2\pi c}T_{wu}\int \rho_w^{G,0}\,\mathrm{d}V}_{\mathrm{SD^{LD}}} \\
&= \underbrace{\frac{\gamma^M}{4\pi c}T_{ku}\left[-2\int \boldsymbol{j}^0_{\mathrm{orb}}\cdot(\boldsymbol{e}_k\times\boldsymbol{r}_G)\,\mathrm{d}V\right]}_{(\gamma^M/4\pi c)T_{ku}g^{\mathrm{OZ}}_{kv}} + \underbrace{\left[-\frac{\gamma^M}{\pi c}\int \boldsymbol{j}^0_{\mathrm{orb}}\cdot\nabla\chi_u\,\mathrm{d}V\right]}_{0} \\
&\quad + \underbrace{\frac{\gamma^M}{4\pi c}T_{wu}\left[2\int \rho_w^{G,0}\,\mathrm{d}V\right]}_{(\gamma^M/4\pi c)T_{wu}g^{\mathrm{SZ}}_{wv}} \\
&= \frac{\gamma^M}{4\pi c}T_{ku}\underbrace{\left[g^{\mathrm{OZ}}_{kv}+g^{\mathrm{SZ}}_{kv}\right]}_{g_{kv}} \\
&= \boxed{\frac{\gamma^M}{4\pi c}T_{ku}(\boldsymbol{R})g_{kv}.}
\end{aligned} \tag{92}$$

16 The result at operator, energy, and spectroscopic levels

Purpose of this section

State the exact same result three times, once per level a reader might actually need it at: as an operator identity, as a differentiated-energy identity, and as the spectroscopic A-vs-g relation that would be compared to experiment.

At operator level,

$$\boxed{\left(\frac{\partial h^D}{\partial \mu_u^M}\right)_{\mathrm{LD}} = T_{ku}(\boldsymbol{R})\frac{\partial h^D}{\partial B_k} + \boldsymbol{\alpha}\cdot\nabla\chi_u.} \tag{93}$$

At the stationary energy level,

$$\boxed{\left.\frac{\mathrm{d}E(J_v)}{\mathrm{d}\mu_u^M}\right|_{\mathrm{LD}} = T_{ku}(\boldsymbol{R})\frac{\mathrm{d}E(J_v)}{\mathrm{d}B_k}.} \tag{94}$$

At the spectroscopic level,

$$\boxed{A^{M,\mathrm{LD}}_{uv} = \frac{\gamma^M}{4\pi c}T_{ku}(\boldsymbol{R})g_{kv}.} \tag{95}$$

17 Comparison with the RKB operator route

Purpose of this table

Line up the RKB route and the Gordon route side by side to confirm they are answering the same question in different languages, and that they agree.

	RKB operator route	**Gordon current route**
Starting object	$\partial h^D/\partial B_k$ and $\partial h^D/\partial \mu_u^M$	The same derivatives
Representation	RKB upper/lower blocks	Current integral
Local orbital term	PSO + relativistic structures	Orbital current $\to$ PSO
Local spin term	Lower-block derivative $\to$ FC+SD	Spin-current integration by parts $\to$ FC+SD
Long-distance FC	No delta derivative	No delta derivative
Long-distance SD	$T_{wu}\sigma_w$	$T_{wu}\rho_w^{G,0}$
Gauge remainder	Commutator $\mathrm{i}[h^{D,(0)}, f]$	Gradient-current integral
Final result	Transferred complete Zeeman matrix element	$A^{M,\mathrm{LD}} = (\gamma^M/4\pi c)Tg$

Normalization note

The RKB operator identity, Webinar-9 tensor formulas, and Debora's tensor formulas use different property normalizations, so their numerical prefactors should not be compared directly. Their common physical content is the same geometric contraction $T(\boldsymbol{R}) \cdot g$.

18 Debora's potential-dependent DKS–Gordon route

Why we redo it a third time

Both routes above quietly assumed the DFT potential V is a plain scalar. **We want to check** whether the long-distance transfer relation $A^{\mathrm{LD}} = (\ldots)Tg$ still holds once we allow V to be a genuine, spin-dependent (non-collinear) exchange–correlation potential — the kind actually used in production relativistic DFT codes.

18.1 Why the two Dirac Hamiltonians look different

Goal of this step

Reconcile the Webinar-9 Hamiltonian with the dissertation's Hamiltonian, which differ by how the rest-mass energy is counted.

What we get: a one-line proof that the difference is a constant shift, so it cannot affect spinors, currents, or any magnetic-derivative operator.

The Webinar Hamiltonian is

$$h^D_{\mathrm{W}} = c\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + \beta c^2 + V_{4\times 4}, \tag{96}$$

whereas the dissertation uses

$$\boxed{h^D_{\mathrm{D}} = c\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + (\beta - \mathbf{1}_4)c^2 + V = h^D_{\mathrm{W}} - c^2\mathbf{1}_4.} \tag{97}$$

This is only a constant rest-energy shift. It changes the orbital energy zero but not the spinors, current, velocity, or magnetic derivative operators.

18.2 Potential-dependent Gordon current

Goal of this step

Find out what a general potential V contributes to the Gordon current — and show that a plain scalar potential contributes nothing extra, while a non-collinear exchange–correlation potential contributes a genuinely new current $\boldsymbol{j}_{\mathrm{xc}}$.

What we get: the extra potential current $\boldsymbol{j}_V$, and the explicit formula for $\boldsymbol{j}_{\mathrm{xc}}$ when it is non-zero.

The general DKS Gordon decomposition at zero external vector potential has the structure

$$\boxed{\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} + \boldsymbol{j}^0_V.} \tag{98}$$

The potential contribution is

$$(j_V)_k = \frac{1}{2c}\left[\phi^\dagger\alpha_k\beta(V\phi) + (V\phi)^\dagger\beta\alpha_k\phi\right]. \tag{99}$$

For a local scalar potential $V_s = v(\boldsymbol{r})\mathbf{1}_4$, this vanishes because

$$\alpha_k\beta + \beta\alpha_k = 0.$$

For a noncollinear xc potential,

$$V^{\mathrm{xc}} = F_l\Sigma_l, \tag{100}$$

one obtains

$$\boxed{\boldsymbol{j}_{\mathrm{xc}} = -\frac{\mathrm{i}}{c}\boldsymbol{F}\times\boldsymbol{Q}, \qquad \boldsymbol{Q} = \phi^\dagger\beta\boldsymbol{\alpha}\phi.} \tag{101}$$

Thus

$$\boxed{\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} + \boldsymbol{j}^0_{\mathrm{xc}}.} \tag{102}$$

Potential part	Extra Gordon current
local scalar nuclear/Coulomb part	0
scalar xc part	0
spin-dependent noncollinear xc part	$\boldsymbol{j}_{\mathrm{xc}} \neq 0$

Detailed derivation: Appendix C.1.

18.3 Debora's g- and HFC decompositions

Goal of this step

State the g-tensor and HFC-tensor decompositions once $\boldsymbol{j}_{\mathrm{xc}}$ is included as a genuine fourth current.

What we get: one extra additive term, xc, alongside OZ/SZ and PSO/FC/SD — no other term changes shape.

In the dissertation normalization,

$$\boxed{g_{uv} = g_{uv}^{\mathrm{OZ}} + g_{uv}^{\mathrm{SZ}} + g_{uv}^{\mathrm{xc}}.} \tag{103}$$

The exact local HFC tensor is

$$\boxed{A_{uv}^{M} = A_{uv}^{M,\mathrm{PSO}} + A_{uv}^{M,\mathrm{FC}} + A_{uv}^{M,\mathrm{SD}} + A_{uv}^{M,\mathrm{xc}}.} \tag{104}$$

The PSO, FC, and SD terms have the same geometric origins as in the scalar-potential Gordon route; the additional term comes from $\boldsymbol{j}_{\mathrm{xc}}$.

18.4 Long-distance reduction of the complete Debora decomposition

Goal of this step

Push every one of the four currents (orb, spin, xc) through the same long-distance gauge identity used in Section 15, and check whether the xc current spoils the clean $T \cdot g$ transfer relation.

What we get: it does not: FC still vanishes, SD still maps cleanly onto $T \cdot g^{\mathrm{SZ}}$, and although PSO and xc individually pick up gauge remainders Δ^{orb}, Δ^{xc}, these two are forced to cancel *exactly* against each other by charge continuity, so the final formula keeps the same clean shape as before.

The same field

$$\boldsymbol{a}^{M,\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_G$$

and the same gauge identity are used. For any partial current X,

$$\boxed{A_{uv}^{M,X,\mathrm{LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^{X} + \Delta_{uv}^{X},} \qquad \Delta_{uv}^{X} = -\frac{\gamma^M}{2c} \int \boldsymbol{j}_X^0 \cdot \nabla \chi_u \, \mathrm{d}V. \tag{105}$$

Therefore

$$A_{uv}^{M,\mathrm{PSO,LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^{\mathrm{OZ}} + \Delta_{uv}^{\mathrm{orb}}, \tag{106}$$

$$A_{uv}^{M,\mathrm{FC,LD}} = 0, \tag{107}$$

$$A_{uv}^{M,\mathrm{SD,LD}} = \frac{\gamma^M}{2c} T_{wu} g_{wv}^{\mathrm{SZ}}, \tag{108}$$

$$A_{uv}^{M,\mathrm{xc,LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^{\mathrm{xc}} + \Delta_{uv}^{\mathrm{xc}}. \tag{109}$$

The spin current is divergence-free identically. Stationary continuity therefore gives

$$\boxed{\Delta_{uv}^{\mathrm{orb}} + \Delta_{uv}^{\mathrm{xc}} = 0.} \tag{110}$$

Hence

$$\boxed{A^{M,\mathrm{LD}}_{uv} = \frac{\gamma^M}{2c} T_{ku}(\boldsymbol{R}) g_{kv}.} \tag{111}$$

18.5 Direct comparison of the two Gordon routes

	Webinar 9 route	**Debora DKS route**
Hamiltonian zero	$c\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + \beta c^2 + V_{4\times 4}$.	$c\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + (\beta - \mathbf{1}_4)c^2 + V$. It differs only by $-c^2\mathbf{1}_4$.
Potential treatment	The displayed decomposition assumes the potential contribution vanishes, as for a scalar Hermitian potential.	The term $\boldsymbol{j}_V$ is retained. Scalar, Coulomb, and occupied-summed HF parts vanish; spin-dependent non-collinear xc leaves $\boldsymbol{j}_{\mathrm{xc}}$.
Field-free current	$\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}}$.	$\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} + \boldsymbol{j}^0_{\mathrm{xc}}$.
Local HFC pieces	PSO, FC, SD.	PSO, FC, SD, and xc HFC.
Long-distance orbital part	PSO maps to the OZ part; in the simplified current the gradient integral vanishes separately.	PSO maps to $(\gamma^M/2c)T\cdot g^{\mathrm{OZ}} + \Delta^{\mathrm{orb}}$.
Long-distance spin part	FC $\to 0$, SD $\to T\cdot g^{\mathrm{SZ}}$.	Exactly the same structural mapping: FC $\to 0$, SD $\to (\gamma^M/2c)T\cdot g^{\mathrm{SZ}}$.
Long-distance xc part	Not present in the displayed Webinar decomposition.	xc HFC $\to (\gamma^M/2c)T\cdot g^{\mathrm{xc}} + \Delta^{\mathrm{xc}}$.
Gauge cancellation	The orbital gradient term is zero for the simplified conserved orbital current.	Only the sum is generally zero: $\Delta^{\mathrm{orb}} + \Delta^{\mathrm{xc}} = 0$.
Final convention	$A^{M,\mathrm{LD}}_{uv} = (\gamma^M/4\pi c)T_{ku}g_{kv}$ in the Webinar nuclear-moment normalization.	$A^{M,\mathrm{LD}}_{uv} = (\gamma^M/2c)T_{ku}g_{kv}$ in the dissertation I^M-derivative normalization.

The factors $1/(4\pi)$ and $1/2$ in the last row must not be compared before converting the definitions of the nuclear perturbation, gyromagnetic ratio, and HFC tensor. The invariant structural statement is identical:

$$\boxed{\text{long-distance HFC} = \text{point-dipole tensor} \times \text{complete Gordon-decomposed } g\text{-tensor.}} \tag{112}$$

In the simplified Webinar form, "complete" means $g^{\mathrm{OZ}} + g^{\mathrm{SZ}}$. In Debora's noncollinear DKS form, it means

$$\boxed{g_{uv} = g^{\mathrm{OZ}}_{uv} + g^{\mathrm{SZ}}_{uv} + g^{\mathrm{xc}}_{uv}.} \tag{113}$$

Complete Debora term-by-term correspondence

$$\begin{aligned}
\mathrm{PSO} &\longrightarrow \frac{\gamma^M}{2c}\boldsymbol{T}\cdot\boldsymbol{g}^{\mathrm{OZ}} + \boldsymbol{\Delta}^{\mathrm{orb}},\\
\mathrm{FC} &\longrightarrow 0,\\
\mathrm{SD} &\longrightarrow \frac{\gamma^M}{2c}\boldsymbol{T}\cdot\boldsymbol{g}^{\mathrm{SZ}},\\
\mathrm{xc}_{\mathrm{HFC}} &\longrightarrow \frac{\gamma^M}{2c}\boldsymbol{T}\cdot\boldsymbol{g}^{\mathrm{xc}} + \boldsymbol{\Delta}^{\mathrm{xc}},\\
\boldsymbol{\Delta}^{\mathrm{orb}} + \boldsymbol{\Delta}^{\mathrm{xc}} &= 0,\\
\boldsymbol{A}^{M,\mathrm{LD}} &= \frac{\gamma^M}{2c}\boldsymbol{T}\cdot(\boldsymbol{g}^{\mathrm{OZ}} + \boldsymbol{g}^{\mathrm{SZ}} + \boldsymbol{g}^{\mathrm{xc}}) = \frac{\gamma^M}{2c}\boldsymbol{T}\cdot\boldsymbol{g}.
\end{aligned}$$

The potential term surviving in Debora's current is the spin-dependent xc contribution. It is not generated by the shifted diagonal blocks of the Hamiltonian. The rest-energy shift changes only the numerical zero of the orbital energies and cancels from the Gordon algebra through $\{\alpha_k, \beta\} = 0$.

Detailed derivation: Appendix C.2.

Complete one-equation map for Debora's full DKS–Gordon decomposition

The chain below starts from the exact local decomposition of Eqs. (4.43) and (4.51) of the dissertation and displays every long-distance image in the same equation.

$$
\boxed{
\begin{aligned}
A^M_{uv} &= \underbrace{\frac{\gamma^M}{2cS}\int \phi_i^{0\dagger} 2\left[\frac{\boldsymbol r_M\times\boldsymbol p}{r_M^3}\right]_u \beta\phi_i^0\,\mathrm dV}_{\text{PSO}\,\leftarrow\,\boldsymbol j^0_{\text{orb}}} + \underbrace{\frac{\gamma^M}{2cS}\int\frac{8\pi}{3}\delta^{(3)}(\boldsymbol r_M)\rho_u^{G,0}\,\mathrm dV}_{\text{FC}\,\leftarrow\,\text{delta part of }\partial_u a^M_w}\\
&\quad+\underbrace{\frac{\gamma^M}{2cS}\int\frac{3(r_M)_u(r_M)_w-\delta_{uw}r_M^2}{r_M^5}\rho_w^{G,0}\,\mathrm dV}_{\text{SD}\,\leftarrow\,\text{regular traceless part of }\partial_u a^M_w}\\
&\quad+\underbrace{\frac{\mathrm i\gamma^M}{c^2S}\int\left[F_u\,\phi_i^{0\dagger}\beta(\boldsymbol a^M\cdot\boldsymbol\alpha)\phi_i^0-(\boldsymbol a^M\cdot\boldsymbol F)\phi_i^{0\dagger}\beta\alpha_u\phi_i^0\right]\mathrm dV}_{\text{xc}_{\text{HFC}}\,\leftarrow\,\boldsymbol j^0_{\text{xc}}=-(\mathrm i/c)\boldsymbol F\times\boldsymbol Q}\\
&\xrightarrow[\partial_u a_w^{M,\text{LD}}=-T_{wu},\ \partial_w a_w^{M,\text{LD}}=0]{\boldsymbol a^M\mapsto\boldsymbol a^{M,\text{LD}}=\boldsymbol R/R^3-\boldsymbol T\boldsymbol r_G,\quad \boldsymbol e_u\times\boldsymbol a^{M,\text{LD}}=\frac12T_{ku}(\boldsymbol e_k\times\boldsymbol r_G)+\nabla\chi_u}\\
A^{M,\text{LD}}_{uv} &= \underbrace{-\frac{\gamma^M}{cS}\int \boldsymbol j^0_{\text{orb}}\cdot\left[\frac12T_{ku}(\boldsymbol e_k\times\boldsymbol r_G)+\nabla\chi_u\right]\mathrm dV}_{\text{PSO}^{\text{LD}}}+\underbrace{0}_{\text{FC}^{\text{LD}}}+\underbrace{\frac{\gamma^M}{2cS}T_{wu}\int\rho_w^{G,0}\,\mathrm dV}_{\text{SD}^{\text{LD}}}\\
&\quad+\underbrace{-\frac{\gamma^M}{cS}\int\boldsymbol j^0_{\text{xc}}\cdot\left[\frac12T_{ku}(\boldsymbol e_k\times\boldsymbol r_G)+\nabla\chi_u\right]\mathrm dV}_{\text{xc}^{\text{LD}}_{\text{HFC}}}\\
&=\underbrace{\frac{\gamma^M}{2c}T_{ku}\left[-\frac1S\int\boldsymbol j^0_{\text{orb}}\cdot(\boldsymbol e_k\times\boldsymbol r_G)\,\mathrm dV\right]}_{(\gamma^M/2c)T_{ku}g^{\text{OZ}}_{kv}}+\underbrace{\left[-\frac{\gamma^M}{cS}\int\boldsymbol j^0_{\text{orb}}\cdot\nabla\chi_u\,\mathrm dV\right]}_{\Delta^{\text{orb}}_{uv}}\\
&\quad+\underbrace{\frac{\gamma^M}{2c}T_{wu}\left[\frac1S\int\rho_w^{G,0}\,\mathrm dV\right]}_{(\gamma^M/2c)T_{wu}g^{\text{SZ}}_{wv}}+\underbrace{\frac{\gamma^M}{2c}T_{ku}\left[-\frac1S\int\boldsymbol j^0_{\text{xc}}\cdot(\boldsymbol e_k\times\boldsymbol r_G)\,\mathrm dV\right]}_{(\gamma^M/2c)T_{ku}g^{\text{xc}}_{kv}}+\underbrace{\left[-\frac{\gamma^M}{cS}\int\boldsymbol j^0_{\text{xc}}\cdot\nabla\chi_u\,\mathrm dV\right]}_{\Delta^{\text{xc}}_{uv}}\\
&=\frac{\gamma^M}{2c}T_{ku}(\boldsymbol R)\left\{\underbrace{\frac1S\int\phi_i^{0\dagger}(\boldsymbol r_G\times\boldsymbol p)_k\beta\phi_i^0\,\mathrm dV+\frac1S\int\rho_k^{G,0}\,\mathrm dV}_{g^{\text{OZ}}_{kv}+g^{\text{SZ}}_{kv}}\right.\\
&\qquad\left.\underbrace{+\frac{\mathrm i}{cS}\int\left[F_k\phi_i^{0\dagger}\beta(\boldsymbol r_G\cdot\boldsymbol\alpha)\phi_i^0-(\boldsymbol r_G\cdot\boldsymbol F)\phi_i^{0\dagger}\beta\alpha_k\phi_i^0\right]\mathrm dV}_{+g^{\text{xc}}_{kv}=g_{kv}}\right\}+\underbrace{\Delta^{\text{orb}}_{uv}+\Delta^{\text{xc}}_{uv}}_{0\text{ by stationary continuity}}\\
&=\boxed{\frac{\gamma^M}{2c}T_{ku}(\boldsymbol R)g_{kv}}.
\end{aligned}}
\tag{114}
$$

19 Final conclusion

What we set out to do — and what we found

We wanted to know what happens to the hyperfine-coupling operator of a nucleus that is far from the electron density we actually compute with. We found that it collapses onto the ordinary Zeeman operator, reshaped by a fixed geometric tensor, with everything else provably vanishing for physical states. The result was checked three independent ways (RKB operator blocks, Gordon current, and Debora's potential-dependent current), and all three agree.

The RKB and Gordon derivations are two representations of the same long-distance physics:

$$\boxed{\text{exact singular HFC field} \;\longrightarrow\; \text{smooth long-distance dipole field} \;\longrightarrow\; T(\boldsymbol{R})\cdot\text{Zeeman response} + \text{gauge remainder.}} \tag{115}$$

The FC term is lost precisely at the singular-to-smooth field replacement. The SD term becomes the transferred spin-Zeeman response. The orbital and relativistic pieces require the gauge rearrangement. For a stationary state the complete gauge remainder vanishes, giving the point-dipole transfer relation between HFC and g.

A Detailed RKB derivations

A.1 Algebraic identities and the distributional Coulomb derivative

← Back to the corresponding main-text step

All Cartesian indices repeated in a single term are summed over x, y, z. We use

$$(\boldsymbol{u} \times \boldsymbol{v})_l = \varepsilon_{lij} u_i v_j, \qquad \boldsymbol{\sigma} \cdot \boldsymbol{p} = \sigma_k p_k, \qquad p_k = -\mathrm{i}\partial_k. \tag{116}$$

The Pauli algebra is

$$\sigma_i \sigma_j = \delta_{ij} + \mathrm{i}\varepsilon_{ijk}\sigma_k, \tag{117}$$

with the Levi–Civita contractions

$$\varepsilon_{ijk}\varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}, \tag{118}$$

$$\varepsilon_{ijk}\varepsilon_{ljk} = 2\delta_{il}. \tag{119}$$

For a coordinate-dependent scalar or Cartesian component $F(\boldsymbol{r})$, the momentum product rule is an operator identity,

$$p_k F = F p_k + [p_k, F] = F p_k - \mathrm{i}\partial_k F. \tag{120}$$

This is the only reason why the lower RKB block later contains derivatives of the hyperfine kernel while the upper block does not.

For the Coulomb kernel the second derivative of $1/r$ has to be understood distributionally. The exact identity is

$$\partial_i \partial_j \frac{1}{r} = \mathrm{PV}\left(\frac{3r_i r_j - r^2\delta_{ij}}{r^5}\right) - \frac{4\pi}{3}\delta_{ij}\delta^{(3)}(\boldsymbol{r}). \tag{121}$$

Since

$$\frac{r_i}{r^3} = -\partial_i \frac{1}{r}, \tag{122}$$

we obtain

$$\partial_j \frac{r_i}{r^3} = -\partial_j \partial_i \frac{1}{r} \tag{123}$$

$$= \mathrm{PV}\left(\frac{\delta_{ij}r^2 - 3r_i r_j}{r^5}\right) + \frac{4\pi}{3}\delta_{ij}\delta^{(3)}(\boldsymbol{r}). \tag{124}$$

Away from the origin the principal-value notation can be suppressed, but it must be retained when the nuclear singularity belongs to the integration domain. The δ-term in Eq. (124) is the unique source of the Fermi-contact term below.

A.2 Component-by-component magnetic derivatives

← Back to the corresponding main-text step

For a parameter λ that enters only through the vector potential, the Dirac Hamiltonian of Eq. (8) gives

$$\frac{\partial h^D}{\partial \lambda} = c \begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \dfrac{\partial \boldsymbol{A}}{\partial \lambda} \\ \boldsymbol{\sigma} \cdot \dfrac{\partial \boldsymbol{A}}{\partial \lambda} & 0 \end{pmatrix}. \tag{125}$$

We now evaluate the Pauli contraction for the two perturbations without skipping the sign changes.

Homogeneous field. With

$$A_j^B = \frac{1}{2}\varepsilon_{jab} B_a r_{O,b}, \tag{126}$$

we have

$$\frac{\partial A_j^B}{\partial B_m} = \frac{1}{2}\varepsilon_{jab}\frac{\partial B_a}{\partial B_m} r_{O,b} = \frac{1}{2}\varepsilon_{jab}\delta_{am} r_{O,b} = \frac{1}{2}\varepsilon_{jmb} r_{O,b}. \tag{127}$$

Therefore

$$\sigma_j \frac{\partial A_j^B}{\partial B_m} = \frac{1}{2}\sigma_j \varepsilon_{jmb} r_{O,b} \tag{128}$$

$$= -\frac{1}{2}\varepsilon_{mjb}\sigma_j r_{O,b} \tag{129}$$

$$= \frac{1}{2}\varepsilon_{mbj} r_{O,b}\sigma_j = \frac{1}{2}(\boldsymbol{r}_O \times \boldsymbol{\sigma})_m. \tag{130}$$

Insertion into Eq. (125) gives Eq. (9).

Nuclear magnetic moment. For nucleus K,

$$A_j^K = \alpha^2 \varepsilon_{jab} M_{K,a} \frac{r_{K,b}}{r_K^3}. \tag{131}$$

Hence

$$\frac{\partial A_j^K}{\partial M_{K,l}} = \alpha^2 \varepsilon_{jab}\delta_{al}\frac{r_{K,b}}{r_K^3} = \alpha^2\varepsilon_{jlb}\frac{r_{K,b}}{r_K^3}, \tag{132}$$

and

$$\sigma_j \frac{\partial A_j^K}{\partial M_{K,l}} = \alpha^2 \sigma_j \varepsilon_{jlb}\frac{r_{K,b}}{r_K^3} \tag{133}$$

$$= -\alpha^2 \varepsilon_{ljb}\sigma_j \frac{r_{K,b}}{r_K^3} \tag{134}$$

$$= \alpha^2 \varepsilon_{lbj}\frac{r_{K,b}}{r_K^3}\sigma_j \tag{135}$$

$$= \alpha^2 \left(\frac{\boldsymbol{r}_K}{r_K^3} \times \boldsymbol{\sigma}\right)_l. \tag{136}$$

This gives Eq. (10). The only structural difference between the two parent operators is therefore the replacement of the smooth coordinate $\boldsymbol{r}_O$ by the singular Coulomb vector field $\boldsymbol{r}_K/r_K^3$.

A.3 Explicit RKB sandwich

← Back to the corresponding main-text step

The RKB embedding is

$$X^{\mathrm{RKB}} = \begin{pmatrix} 1 & 0 \\ 0 & \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{2c} \end{pmatrix}. \tag{137}$$

On the chosen operator domain $(\boldsymbol{\sigma}\cdot\boldsymbol{p})^\dagger = \boldsymbol{\sigma}\cdot\boldsymbol{p}$; in AO matrix elements this is the usual integration-by-parts statement with vanishing boundary terms.

For

$$\mathcal{O}[Q,\lambda] = \lambda\begin{pmatrix} 0 & Q \\ Q & 0 \end{pmatrix}, \tag{138}$$

first multiply on the right,

$$\mathcal{O}[Q,\lambda]X^{\mathrm{RKB}} = \lambda\begin{pmatrix} 0 & Q \\ Q & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & (\boldsymbol{\sigma}\cdot\boldsymbol{p})/(2c) \end{pmatrix} \tag{139}$$

$$= \lambda \begin{pmatrix} 0 & Q(\boldsymbol{\sigma}\cdot\boldsymbol{p})/(2c) \\ Q & 0 \end{pmatrix}. \tag{140}$$

Multiplication from the left then gives

$$(X^{\mathrm{RKB}})^\dagger \mathcal{O}[Q,\lambda] X^{\mathrm{RKB}} = \begin{pmatrix} 1 & 0 \\ 0 & (\boldsymbol{\sigma}\cdot\boldsymbol{p})/(2c) \end{pmatrix} \lambda \begin{pmatrix} 0 & Q(\boldsymbol{\sigma}\cdot\boldsymbol{p})/(2c) \\ Q & 0 \end{pmatrix} \tag{141}$$

$$= \frac{\lambda}{2c} \begin{pmatrix} 0 & Q\boldsymbol{\sigma}\cdot\boldsymbol{p} \\ \boldsymbol{\sigma}\cdot\boldsymbol{p}\, Q & 0 \end{pmatrix}. \tag{142}$$

Thus

$$U[Q] = Q\boldsymbol{\sigma}\cdot\boldsymbol{p}, \qquad L[Q] = \boldsymbol{\sigma}\cdot\boldsymbol{p}\, Q. \tag{143}$$

The asymmetry is essential: in $L[Q]$ the momentum differentiates every coordinate-dependent factor in Q before acting on the wavefunction to the right.

A.4 Generic upper and lower RKB blocks

← Back to the corresponding main-text step

For a general vector field $\boldsymbol{a}(\boldsymbol{r})$, define

$$Q_l[\boldsymbol{a}] = (\boldsymbol{a}\times\boldsymbol{\sigma})_l = \varepsilon_{lij} a_i \sigma_j. \tag{144}$$

Upper block. Because the momentum is already to the right of Q_l, it does not differentiate a_i:

$$U_l[\boldsymbol{a}] = Q_l[\boldsymbol{a}]\, \boldsymbol{\sigma}\cdot\boldsymbol{p} \tag{145}$$

$$= \varepsilon_{lij} a_i \sigma_j \sigma_k p_k \tag{146}$$

$$= \varepsilon_{lij} a_i (\delta_{jk} + \mathrm{i}\varepsilon_{jkn}\sigma_n) p_k \tag{147}$$

$$= \varepsilon_{lik} a_i p_k + \mathrm{i}\varepsilon_{lij}\varepsilon_{jkn} a_i p_k \sigma_n \tag{148}$$

$$= (\boldsymbol{a}\times\boldsymbol{p})_l + \mathrm{i}[\boldsymbol{a}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l. \tag{149}$$

Using $\varepsilon_{lij}\varepsilon_{jkn} = \delta_{lk}\delta_{in} - \delta_{ln}\delta_{ik}$, the second term may also be written as

$$[\boldsymbol{a}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l = (\boldsymbol{a}\cdot\boldsymbol{\sigma}) p_l - (\boldsymbol{a}\cdot\boldsymbol{p})\sigma_l. \tag{150}$$

Lower block. Now the product rule is unavoidable:

$$L_l[\boldsymbol{a}] = \boldsymbol{\sigma}\cdot\boldsymbol{p}\, Q_l[\boldsymbol{a}] \tag{151}$$

$$= \sigma_k p_k\, \varepsilon_{lij} a_i \sigma_j \tag{152}$$

$$= \varepsilon_{lij}\sigma_k (a_i p_k + [p_k, a_i])\sigma_j \tag{153}$$

$$= \underbrace{\varepsilon_{lij} a_i \sigma_k \sigma_j p_k}_{\text{non-derivative part}} + \underbrace{\varepsilon_{lij}\sigma_k [p_k, a_i]\sigma_j}_{D_l[\boldsymbol{a}]}. \tag{154}$$

In the first term the Pauli matrices occur in the reversed order,

$$\varepsilon_{lij} a_i \sigma_k \sigma_j p_k = \varepsilon_{lij} a_i (\delta_{kj} + \mathrm{i}\varepsilon_{kjn}\sigma_n) p_k \tag{155}$$

$$= (\boldsymbol{a}\times\boldsymbol{p})_l - \mathrm{i}[\boldsymbol{a}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l, \tag{156}$$

because $\varepsilon_{kjn} = -\varepsilon_{jkn}$.

For the derivative part,

$$D_l[\boldsymbol{a}] = \varepsilon_{lij}\sigma_k(-\mathrm{i}\partial_k a_i)\sigma_j \tag{157}$$

$$= -\mathrm{i}\varepsilon_{lij}(\partial_k a_i)(\delta_{kj} + \mathrm{i}\varepsilon_{kjn}\sigma_n) \tag{158}$$

$$= -\mathrm{i}\varepsilon_{lik}\partial_k a_i + \varepsilon_{lij}\varepsilon_{kjn}(\partial_k a_i)\sigma_n. \tag{159}$$

Using

$$\varepsilon_{lij}\varepsilon_{kjn} = \delta_{ln}\delta_{ik} - \delta_{lk}\delta_{in}, \tag{160}$$

this becomes

$$D_l[\boldsymbol{a}] = \mathrm{i}\varepsilon_{lki}\partial_k a_i + (\partial_i a_i)\sigma_l - (\partial_l a_n)\sigma_n \tag{161}$$
$$= \mathrm{i}(\nabla \times \boldsymbol{a})_l + (\nabla \cdot \boldsymbol{a})\sigma_l - (\partial_l a_m)\sigma_m. \tag{162}$$

Therefore

$$\boxed{L_l[\boldsymbol{a}] = (\boldsymbol{a} \times \boldsymbol{p})_l - \mathrm{i}[\boldsymbol{a} \times (\boldsymbol{p} \times \boldsymbol{\sigma})]_l + \mathrm{i}(\nabla \times \boldsymbol{a})_l + (\nabla \cdot \boldsymbol{a})\sigma_l - (\partial_l a_m)\sigma_m.} \tag{163}$$

This is Eq. (16); it is evaluated twice below, first for the exact singular field and then for the smooth long-distance field.

A.5 Distributional derivation of FC and SD

← Back to the corresponding main-text step

Set temporarily $r_m \equiv r_{K,m}$ and

$$a_m^K = \frac{r_m}{r^3}. \tag{164}$$

Equation (124) gives

$$\partial_l a_m^K = \mathrm{PV}\left(\frac{\delta_{lm} r^2 - 3r_l r_m}{r^5}\right) + \frac{4\pi}{3}\delta_{lm}\delta^{(3)}(\boldsymbol{r}). \tag{165}$$

Taking the trace over $l = m$,

$$\nabla \cdot \boldsymbol{a}^K = \partial_m a_m^K \tag{166}$$
$$= \mathrm{PV}\left(\frac{3r^2 - 3r_m r_m}{r^5}\right) + \frac{4\pi}{3}(3)\delta^{(3)}(\boldsymbol{r}) \tag{167}$$
$$= 4\pi\delta^{(3)}(\boldsymbol{r}). \tag{168}$$

The curl vanishes distributionally because every tensor in Eq. (165) is symmetric in l, m:

$$(\nabla \times \boldsymbol{a}^K)_q = \varepsilon_{qlm}\partial_l a_m^K = 0. \tag{169}$$

Insert Eqs. (165) and (168) into the derivative sector of the lower block:

$$(\nabla \cdot \boldsymbol{a}^K)\sigma_l - (\partial_l a_m^K)\sigma_m \tag{170}$$
$$= 4\pi\delta^{(3)}(\boldsymbol{r}_K)\sigma_l - \left[\mathrm{PV}\left(\frac{\delta_{lm} r_K^2 - 3r_{K,l} r_{K,m}}{r_K^5}\right) + \frac{4\pi}{3}\delta_{lm}\delta^{(3)}(\boldsymbol{r}_K)\right]\sigma_m \tag{171}$$
$$= 4\pi\delta^{(3)}(\boldsymbol{r}_K)\sigma_l - \frac{4\pi}{3}\delta^{(3)}(\boldsymbol{r}_K)\sigma_l + \mathrm{PV}\left(\frac{3r_{K,l}(\boldsymbol{r}_K \cdot \boldsymbol{\sigma}) - r_K^2\sigma_l}{r_K^5}\right) \tag{172}$$
$$= \boxed{\frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_K)\sigma_l + \mathrm{PV}\left(\frac{3r_{K,l}(\boldsymbol{r}_K \cdot \boldsymbol{\sigma}) - r_K^2\sigma_l}{r_K^5}\right)}. \tag{173}$$

The two terms are exactly FC and SD. Together with

$$\frac{1}{2}\mathrm{PSO}_l = (\boldsymbol{a}^K \times \boldsymbol{p})_l = \frac{(\boldsymbol{r}_K \times \boldsymbol{p})_l}{r_K^3}, \tag{174}$$
$$a\mathrm{REL}_l = \mathrm{i}[\boldsymbol{a}^K \times (\boldsymbol{p} \times \boldsymbol{\sigma})]_l = \frac{\mathrm{i}}{r_K^3}[(\boldsymbol{r}_K \cdot \boldsymbol{\sigma})p_l - (\boldsymbol{r}_K \cdot \boldsymbol{p})\sigma_l], \tag{175}$$

we recover, without approximation,

$$U_l^K = \frac{1}{2}\mathrm{PSO}_l + a\mathrm{REL}_l, \tag{176}$$
$$L_l^K = \frac{1}{2}\mathrm{PSO}_l + \mathrm{FC}_l + \mathrm{SD}_l - a\mathrm{REL}_l. \tag{177}$$

The contact coefficient $8\pi/3$ comes from the combination $4\pi - 4\pi/3$, not from a convention inserted by hand.

A.6 Long-distance Taylor expansion

← Back to the corresponding main-text step

Start from

$$\boldsymbol{a}^K(\boldsymbol{r}) = \frac{\boldsymbol{R}+\boldsymbol{r}_O}{|\boldsymbol{R}+\boldsymbol{r}_O|^3}. \tag{178}$$

The denominator obeys

$$|\boldsymbol{R}+\boldsymbol{r}_O|^2 = (\boldsymbol{R}+\boldsymbol{r}_O)\cdot(\boldsymbol{R}+\boldsymbol{r}_O) = R^2 + 2\boldsymbol{R}\cdot\boldsymbol{r}_O + r_O^2, \tag{179}$$

so

$$|\boldsymbol{R}+\boldsymbol{r}_O|^{-3} = R^{-3}\left(1+\frac{2\boldsymbol{R}\cdot\boldsymbol{r}_O + r_O^2}{R^2}\right)^{-3/2}. \tag{180}$$

With

$$(1+x)^{-3/2} = 1 - \frac{3}{2}x + \mathcal{O}(x^2), \tag{181}$$

and keeping only terms linear in $\boldsymbol{r}_O$,

$$\begin{aligned} |\boldsymbol{R}+\boldsymbol{r}_O|^{-3} &= \frac{1}{R^3}\left[1 - \frac{3}{2}\frac{2\boldsymbol{R}\cdot\boldsymbol{r}_O + r_O^2}{R^2} + \mathcal{O}\left(\frac{r_O^2}{R^2}\right)\right] && (182)\\ &= \frac{1}{R^3} - 3\frac{\boldsymbol{R}\cdot\boldsymbol{r}_O}{R^5} + \mathcal{O}\left(\frac{r_O^2}{R^5}\right). && (183)\end{aligned}$$

Multiplication by $\boldsymbol{R}+\boldsymbol{r}_O$ gives

$$\begin{aligned} \frac{\boldsymbol{R}+\boldsymbol{r}_O}{|\boldsymbol{R}+\boldsymbol{r}_O|^3} &= (\boldsymbol{R}+\boldsymbol{r}_O)\left[\frac{1}{R^3} - 3\frac{\boldsymbol{R}\cdot\boldsymbol{r}_O}{R^5}\right] + \mathcal{O}\left(\frac{r_O^2}{R^4}\right) && (184)\\ &= \frac{\boldsymbol{R}}{R^3} + \frac{\boldsymbol{r}_O}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{R}}{R^5} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{r}_O}{R^5} + \mathcal{O}\left(\frac{r_O^2}{R^4}\right). && (185)\end{aligned}$$

The fourth displayed term is already quadratic in $\boldsymbol{r}_O$ and is absorbed into the remainder. Therefore

$$\boxed{\boldsymbol{a}^{\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} + \frac{\boldsymbol{r}_O}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{R}}{R^5} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_O.} \tag{186}$$

Componentwise,

$$\begin{aligned} a_m^{\mathrm{LD}} &= \frac{R_m}{R^3} + \left(\frac{\delta_{mq}}{R^3} - \frac{3R_mR_q}{R^5}\right) r_{O,q} && (187)\\ &= \frac{R_m}{R^3} - T_{mq}r_{O,q}. && (188)\end{aligned}$$

Since $\boldsymbol{R}$ and T_{mq} are constant with respect to $\boldsymbol{r}_O$,

$$\partial_l a_m^{\mathrm{LD}} = -T_{mq}\partial_l r_{O,q} = -T_{mq}\delta_{lq} = -T_{ml}. \tag{189}$$

The tensor T is symmetric and traceless, hence

$$\begin{aligned} (\nabla\times\boldsymbol{a}^{\mathrm{LD}})_k &= \varepsilon_{klm}\partial_l a_m^{\mathrm{LD}} = -\varepsilon_{klm}T_{ml} = 0, && (190)\\ \nabla\cdot\boldsymbol{a}^{\mathrm{LD}} &= \partial_m a_m^{\mathrm{LD}} = -T_{mm} = 0. && (191)\end{aligned}$$

The remaining derivative structure is

$$-(\partial_l a_m^{\mathrm{LD}})\sigma_m = T_{ml}\sigma_m. \tag{192}$$

Thus the exact singular derivative sector FC + SD is replaced by the smooth term $T_{ml}\sigma_m$. No finite Taylor polynomial about O can recreate the delta distribution supported at the distant point K.

A.7 Detailed expansion of the long-distance RKB blocks

← Back to the corresponding main-text step

Write the retained field as three pieces,

$$\boldsymbol{a}^{\mathrm{LD}} = \boldsymbol{a}^{(0)} + \boldsymbol{a}^{(1a)} + \boldsymbol{a}^{(1b)}, \tag{193}$$

with

$$\boldsymbol{a}^{(0)} = \frac{\boldsymbol{R}}{R^3}, \qquad \boldsymbol{a}^{(1a)} = \frac{\boldsymbol{r}_O}{R^3}, \qquad \boldsymbol{a}^{(1b)} = -3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{R}}{R^5}. \tag{194}$$

Linearity of Eq. (149) gives

$$U_l^{\mathrm{LD}} = (\boldsymbol{a}^{\mathrm{LD}} \times \boldsymbol{p})_l + \mathrm{i}[\boldsymbol{a}^{\mathrm{LD}} \times (\boldsymbol{p}\times\boldsymbol{\sigma})]_l \tag{195}$$

$$= \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} \tag{196}$$

$$+ \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5}. \tag{197}$$

This is Eq. (31).

For the lower block, begin from the full generic identity rather than changing signs by inspection:

$$L_l^{\mathrm{LD}} = (\boldsymbol{a}^{\mathrm{LD}} \times \boldsymbol{p})_l - \mathrm{i}[\boldsymbol{a}^{\mathrm{LD}} \times (\boldsymbol{p}\times\boldsymbol{\sigma})]_l \tag{198}$$

$$+ \mathrm{i}(\nabla\times\boldsymbol{a}^{\mathrm{LD}})_l + (\nabla\cdot\boldsymbol{a}^{\mathrm{LD}})\sigma_l - (\partial_l a_m^{\mathrm{LD}})\sigma_m. \tag{199}$$

The first two derivative terms vanish and the last one is $T_{ml}\sigma_m$. Therefore

$$L_l^{\mathrm{LD}} = \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{(\boldsymbol{r}_O\times\boldsymbol{p})_l}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} \tag{200}$$

$$- \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - \frac{\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} \tag{201}$$

$$+ \sum_m T_{ml}\sigma_m, \tag{202}$$

which is Eq. (32). The last line is the complete derivative contribution of the retained smooth field; there is no hidden contact term left to cancel.

A.8 Origin of the $2\sigma_m$ term in the Zeeman block

← Back to the corresponding main-text step

For

$$Q_m^B = \varepsilon_{mij} r_{O,i}\sigma_j, \tag{203}$$

the upper block is obtained directly from Eq. (149). In the lower block the derivative of $r_{O,i}$ produces an additional term,

$$D_m^B = -\mathrm{i}\varepsilon_{mij}(\partial_k r_{O,i})\sigma_k\sigma_j \tag{204}$$

$$= -\mathrm{i}\varepsilon_{mij}\delta_{ki}\sigma_k\sigma_j \tag{205}$$

$$= -\mathrm{i}\varepsilon_{mij}\sigma_i\sigma_j \tag{206}$$

$$= -\mathrm{i}\varepsilon_{mij}(\delta_{ij} + \mathrm{i}\varepsilon_{ijn}\sigma_n) \tag{207}$$

$$= \varepsilon_{mij}\varepsilon_{ijn}\sigma_n \tag{208}$$

$$= 2\delta_{mn}\sigma_n = 2\sigma_m. \tag{209}$$

Hence

$$U_m^B = (\boldsymbol{r}_O\times\boldsymbol{p})_m + \mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_m, \tag{210}$$

$$L_m^B = (\boldsymbol{r}_O\times\boldsymbol{p})_m - \mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_m + 2\sigma_m, \tag{211}$$

which gives Eq. (35).

A.9 Gauge function, gradient, and direct gauge-block derivation

← Back to the corresponding main-text step

We first make the kernel decomposition explicit. From

$$T_{ml} = \frac{3R_m R_l - R^2\delta_{ml}}{R^5}, \qquad Q_m^B = (\boldsymbol{r}_O \times \boldsymbol{\sigma})_m, \tag{212}$$

one obtains

$$\sum_m T_{ml} Q_m^B = \frac{3R_l}{R^5}\,\boldsymbol{R}\cdot(\boldsymbol{r}_O \times \boldsymbol{\sigma}) - \frac{(\boldsymbol{r}_O \times \boldsymbol{\sigma})_l}{R^3}. \tag{213}$$

Subtracting one half of Eq. (213) from $Q_l^{K,\mathrm{LD}}$ and rearranging the scalar triple products gives exactly the remainder Q_l^G in Eq. (37). We now derive that remainder independently from a scalar gauge function.

Define

$$F_l \equiv \frac{f_l}{\alpha^2}, \qquad B \equiv \boldsymbol{R}\cdot\boldsymbol{r}_O, \qquad C_l \equiv (\boldsymbol{R}\times\boldsymbol{r}_O)_l, \qquad \boldsymbol{c}_l \equiv \boldsymbol{e}_l \times \boldsymbol{R}. \tag{214}$$

Because

$$C_l = (\boldsymbol{e}_l \times \boldsymbol{R})\cdot\boldsymbol{r}_O = \boldsymbol{c}_l\cdot\boldsymbol{r}_O, \tag{215}$$

we have

$$\nabla B = \boldsymbol{R}, \qquad \nabla C_l = \boldsymbol{c}_l. \tag{216}$$

The function

$$F_l = \frac{C_l}{R^3} - \frac{3BC_l}{2R^5} \tag{217}$$

therefore satisfies

$$\nabla F_l = \frac{\boldsymbol{c}_l}{R^3} - \frac{3}{2R^5}\nabla(BC_l) \tag{218}$$

$$= \frac{\boldsymbol{c}_l}{R^3} - \frac{3}{2R^5}(C_l\boldsymbol{R} + B\boldsymbol{c}_l). \tag{219}$$

Contracting with $\boldsymbol{\sigma}$,

$$\alpha^2\boldsymbol{\sigma}\cdot\nabla F_l = \alpha^2\left[\frac{(\boldsymbol{R}\times\boldsymbol{\sigma})_l}{R^3} - \frac{3}{2R^5}\left((\boldsymbol{\sigma}\cdot\boldsymbol{R})(\boldsymbol{R}\times\boldsymbol{r}_O)_l + (\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{\sigma})_l\right)\right] \tag{220}$$

$$= Q_l^G. \tag{221}$$

Thus $Q_l^G = \boldsymbol{\sigma}\cdot\nabla f_l$, as stated in the main text.

Since $[p_k, f_l] = -\mathrm{i}\partial_k f_l$ and the local multiplicative potential and rest-energy blocks commute with f_l,

$$\mathrm{i}[h^{D,(0)}, f_l] = c\begin{pmatrix} 0 & \boldsymbol{\sigma}\cdot\nabla f_l \\ \boldsymbol{\sigma}\cdot\nabla f_l & 0 \end{pmatrix} \tag{222}$$

$$= c\begin{pmatrix} 0 & Q_l^G \\ Q_l^G & 0 \end{pmatrix}. \tag{223}$$

This proves Eq. (40) before any RKB transformation.

Direct upper gauge block. Set $\boldsymbol{g}_l \equiv \nabla F_l$. Then

$$\alpha^{-2}U_l^G = (\boldsymbol{\sigma}\cdot\boldsymbol{g}_l)(\boldsymbol{\sigma}\cdot\boldsymbol{p}) \tag{224}$$

$$= \boldsymbol{g}_l\cdot\boldsymbol{p} + \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{g}_l\times\boldsymbol{p}). \tag{225}$$

The orbital factor is

$$\boldsymbol{g}_l\cdot\boldsymbol{p} = \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} - \frac{3}{2R^5}\left[(\boldsymbol{R}\times\boldsymbol{r}_O)_l(\boldsymbol{R}\cdot\boldsymbol{p}) + (\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l\right]. \tag{226}$$

Use the ordered vector identity

$$(\boldsymbol{R}\times\boldsymbol{r}_O)_l(\boldsymbol{R}\cdot\boldsymbol{X}) = (\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{X})_l + R_l\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{X}) - R^2(\boldsymbol{r}_O\times\boldsymbol{X})_l, \tag{227}$$

valid for any $\boldsymbol{X}$ whose operator ordering relative to $\boldsymbol{r}_O$ is left unchanged. With $\boldsymbol{X} = \boldsymbol{p}$,

$$\boldsymbol{g}_l \cdot \boldsymbol{p} = \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{3(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p}). \tag{228}$$

For the spin-dependent factor note

$$\boldsymbol{\sigma}\cdot[(\boldsymbol{e}_l\times\boldsymbol{R})\times\boldsymbol{p}] = (\boldsymbol{\sigma}\cdot\boldsymbol{R})p_l - \sigma_l(\boldsymbol{R}\cdot\boldsymbol{p}) = [\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l, \tag{229}$$

$$\boldsymbol{\sigma}\cdot(\boldsymbol{R}\times\boldsymbol{p}) = \boldsymbol{R}\cdot(\boldsymbol{p}\times\boldsymbol{\sigma}). \tag{230}$$

Applying Eq. (227) with $\boldsymbol{X} = \boldsymbol{p}\times\boldsymbol{\sigma}$ gives

$$\boldsymbol{\sigma}\cdot(\boldsymbol{g}_l\times\boldsymbol{p}) = \frac{[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{3[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} \tag{231}$$

$$- 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]. \tag{232}$$

Combining Eqs. (225), (228), and (232),

$$\alpha^{-2}U_l^G = \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{3(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p}) \tag{233}$$

$$+ \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} + \frac{3\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} - 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} \tag{234}$$

$$- \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]. \tag{235}$$

Direct lower gauge block. Now the momentum differentiates $\boldsymbol{g}_l$:

$$\alpha^{-2}L_l^G = \boldsymbol{\sigma}\cdot\boldsymbol{p}\,(\boldsymbol{\sigma}\cdot\boldsymbol{g}_l) \tag{236}$$

$$= \boldsymbol{g}_l\cdot\boldsymbol{p} - \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{g}_l\times\boldsymbol{p}) - \mathrm{i}\nabla\cdot\boldsymbol{g}_l + \boldsymbol{\sigma}\cdot(\nabla\times\boldsymbol{g}_l). \tag{237}$$

Because $\boldsymbol{g}_l = \nabla F_l$,

$$\nabla\times\boldsymbol{g}_l = 0. \tag{238}$$

Furthermore B and C_l are linear in $\boldsymbol{r}_O$, so

$$\nabla^2 B = \nabla^2 C_l = 0, \qquad \nabla B\cdot\nabla C_l = \boldsymbol{R}\cdot(\boldsymbol{e}_l\times\boldsymbol{R}) = 0. \tag{239}$$

Hence

$$\nabla^2(BC_l) = B\nabla^2 C_l + C_l\nabla^2 B + 2\nabla B\cdot\nabla C_l = 0, \tag{240}$$

and therefore

$$\nabla\cdot\boldsymbol{g}_l = \nabla^2 F_l = 0. \tag{241}$$

Equation (237) reduces to the same orbital part as U_l^G, but with the opposite sign of the relativistic part:

$$\alpha^{-2}L_l^G = \frac{(\boldsymbol{R}\times\boldsymbol{p})_l}{R^3} + \frac{3(\boldsymbol{r}_O\times\boldsymbol{p})_l}{2R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)(\boldsymbol{R}\times\boldsymbol{p})_l}{R^5} - \frac{3R_l}{2R^5}\boldsymbol{R}\cdot(\boldsymbol{r}_O\times\boldsymbol{p}) \tag{242}$$

$$- \frac{\mathrm{i}[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^3} - \frac{3\mathrm{i}[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{2R^3} + 3\mathrm{i}\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)[\boldsymbol{R}\times(\boldsymbol{p}\times\boldsymbol{\sigma})]_l}{R^5} \tag{243}$$

$$+ \frac{3\mathrm{i}R_l}{2R^5}\boldsymbol{R}\cdot[\boldsymbol{r}_O\times(\boldsymbol{p}\times\boldsymbol{\sigma})]. \tag{244}$$

Equations (235) and (244) are obtained directly from $f_l \to \nabla f_l \to Q_l^G \to (U_l^G, L_l^G)$, not by subtracting the Zeeman blocks from the long-distance hyperfine blocks. They are therefore an independent check of the large term-by-term map in the main text. Because $\nabla\cdot\boldsymbol{g}_l = \nabla\times\boldsymbol{g}_l = 0$, the gauge block generates no additional $T_{ml}\sigma_m$ term: the direct SD $\to T\cdot$SZ mapping stays entirely in the transferred Zeeman sector.

Finally, applying the generic RKB sandwich to the four-component commutator, whose parent prefactor is $\lambda = c$, gives

$$(X^{\mathrm{RKB}})^\dagger \mathrm{i}[h^{D,(0)}, f_l]X^{\mathrm{RKB}} = \frac{1}{2}\begin{pmatrix} 0 & U_l^G \\ L_l^G & 0 \end{pmatrix}, \tag{245}$$

which is Eq. (41).

A.10 Three proofs of the stationary gauge cancellation

← Back to the corresponding main-text step

The gauge operator is generally nonzero. What vanishes is its complete diagonal matrix element in a stationary state; neither U_l^G nor L_l^G is zero separately.

1. Eigenstate proof. Let

$$h^{D,(0)}|\phi_i^0\rangle = \varepsilon_i|\phi_i^0\rangle, \qquad \langle\phi_i^0|h^{D,(0)} = \varepsilon_i\langle\phi_i^0|. \tag{246}$$

Then

$$\begin{aligned} G_{l,i} &\equiv \langle\phi_i^0|\mathrm{i}[h^{D,(0)}, f_l]|\phi_i^0\rangle & (247)\\ &= \mathrm{i}\langle\phi_i^0|h^{D,(0)} f_l|\phi_i^0\rangle - \mathrm{i}\langle\phi_i^0|f_l h^{D,(0)}|\phi_i^0\rangle & (248)\\ &= \mathrm{i}\varepsilon_i\langle\phi_i^0|f_l|\phi_i^0\rangle - \mathrm{i}\varepsilon_i\langle\phi_i^0|f_l|\phi_i^0\rangle & (249)\\ &= 0. & (250)\end{aligned}$$

For completeness, off-diagonal matrix elements need not vanish:

$$\langle\phi_m^0|\mathrm{i}[h^{D,(0)}, f_l]|\phi_n^0\rangle = \mathrm{i}(\varepsilon_m - \varepsilon_n)\langle\phi_m^0|f_l|\phi_n^0\rangle. \tag{251}$$

Thus the cancellation is a stationary-property statement, not an operator identity $\mathrm{i}[h, f_l] = 0$.

2. Density-matrix proof. For

$$D^0 = \sum_{i\in\mathrm{occ}} n_i|\phi_i^0\rangle\langle\phi_i^0|, \tag{252}$$

constructed from eigenstates of the same field-free Hamiltonian,

$$[D^0, h^{D,(0)}] = 0. \tag{253}$$

Hence

$$\begin{aligned} \mathrm{Tr}[D^0\,\mathrm{i}[h^{D,(0)}, f_l]] &= \mathrm{i}\mathrm{Tr}(D^0 h^{D,(0)} f_l - D^0 f_l h^{D,(0)}) & (254)\\ &= \mathrm{i}\mathrm{Tr}[(D^0 h^{D,(0)} - h^{D,(0)} D^0) f_l] & (255)\\ &= \mathrm{i}\mathrm{Tr}([D^0, h^{D,(0)}] f_l) = 0. & (256)\end{aligned}$$

Only cyclicity of the trace is used in the second line.

3. Current-density proof. For the local field-free Hamiltonian,

$$\mathrm{i}[h^{D,(0)}, f_l] = c\boldsymbol{\alpha}\cdot\nabla f_l. \tag{257}$$

With

$$\boldsymbol{j}^0 = -c\phi_i^{0\dagger}\boldsymbol{\alpha}\phi_i^0, \tag{258}$$

the expectation value is

$$\begin{aligned} G_l &= \int \phi_i^{0\dagger}(c\boldsymbol{\alpha}\cdot\nabla f_l)\phi_i^0\,\mathrm{d}V & (259)\\ &= -\int \boldsymbol{j}^0\cdot\nabla f_l\,\mathrm{d}V. & (260)\end{aligned}$$

Using

$$\nabla\cdot(f_l\boldsymbol{j}^0) = \boldsymbol{j}^0\cdot\nabla f_l + f_l\nabla\cdot\boldsymbol{j}^0, \tag{261}$$

we obtain

$$\int \boldsymbol{j}^0\cdot\nabla f_l\,\mathrm{d}V = \oint_{\partial\Omega} f_l\boldsymbol{j}^0\cdot\mathrm{d}\boldsymbol{S} - \int f_l\nabla\cdot\boldsymbol{j}^0\,\mathrm{d}V. \tag{262}$$

The surface term vanishes for a localized bound state. Stationarity gives

$$\partial_t\rho + \nabla\cdot\boldsymbol{j}^0 = 0, \qquad \partial_t\rho = 0 \quad\Longrightarrow\quad \nabla\cdot\boldsymbol{j}^0 = 0. \tag{263}$$

Thus $G_l = 0$. The current proof is the direct bridge to the Gordon derivation in Appendix B.

A.11 AO density contraction

← Back to the corresponding main-text step

Let the oriented molecular spinor be expanded as

$$\phi^0_{m,i}(J_v) = X^{\mathrm{RKB}}_{m,a\kappa} C^0_{a\kappa,i}(J_v). \tag{264}$$

For a one-electron RKB-transformed operator $\widetilde{\mathcal{O}}$,

$$\langle \phi^0_i | \widetilde{\mathcal{O}} | \phi^0_i \rangle = [C^0_{a\kappa,i}(J_v)]^* C^0_{b\eta,i}(J_v) \int \chi^*_\kappa \, \widetilde{\mathcal{O}}_{ab} \, \chi_\eta \, \mathrm{d}V. \tag{265}$$

Relabel the two dummy composite indices $(a\kappa) \leftrightarrow (b\eta)$ and define

$$D^0_{a\kappa,b\eta}(J_v) = C^0_{a\kappa,i}(J_v)[C^0_{b\eta,i}(J_v)]^*. \tag{266}$$

Then

$$\boxed{\langle \phi^0_i | \widetilde{\mathcal{O}} | \phi^0_i \rangle = D^0_{a\kappa,b\eta}(J_v) \int \chi^*_\eta \widetilde{\mathcal{O}}_{ba} \chi_\kappa \, \mathrm{d}V,} \tag{267}$$

which is Eq. (45). No physics is introduced by this step; it is only the AO-index form of the spinor matrix element.

B Detailed Gordon derivations

B.1 Magnetic derivatives in the Webinar-9 convention

← Back to the corresponding main-text step

The four-component Hamiltonian is written as

$$h^D(\lambda) = c\boldsymbol{\alpha} \cdot \boldsymbol{p} + \boldsymbol{\alpha} \cdot \boldsymbol{A}(\lambda) + c^2\beta + V_{4\times4}. \tag{268}$$

If λ enters only through $\boldsymbol{A}$, then

$$\frac{\partial h^D}{\partial \lambda} = \boldsymbol{\alpha} \cdot \frac{\partial \boldsymbol{A}}{\partial \lambda}. \tag{269}$$

For the homogeneous field,

$$\boldsymbol{A}^B = \frac{1}{2}\boldsymbol{B} \times \boldsymbol{r}_G, \qquad A^B_j = \frac{1}{2}\varepsilon_{jab} B_a (r_G)_b. \tag{270}$$

Therefore

$$\frac{\partial A^B_j}{\partial B_u} = \frac{1}{2}\varepsilon_{jub}(r_G)_b, \tag{271}$$

and

$$\begin{aligned} \boldsymbol{\alpha} \cdot \frac{\partial \boldsymbol{A}^B}{\partial B_u} &= \frac{1}{2}\alpha_j \varepsilon_{jub}(r_G)_b & (272)\\ &= \frac{1}{2}(\boldsymbol{r}_G \times \boldsymbol{\alpha})_u. & (273) \end{aligned}$$

Thus

$$\boxed{\frac{\partial h^D}{\partial B_u} = \frac{1}{2}(\boldsymbol{r}_G \times \boldsymbol{\alpha})_u.} \tag{274}$$

For the nuclear moment,

$$\boldsymbol{A}^M = \boldsymbol{\mu}^M \times \frac{\boldsymbol{r}_M}{r^3_M}, \qquad A^M_j = \varepsilon_{jab} \mu^M_a \frac{(r_M)_b}{r^3_M}. \tag{275}$$

Hence

$$\frac{\partial A_j^M}{\partial \mu_u^M} = \varepsilon_{jub} \frac{(r_M)_b}{r_M^3}, \tag{276}$$

and the scalar-triple-product identity gives

$$\boldsymbol{\alpha} \cdot \frac{\partial \boldsymbol{A}^M}{\partial \mu_u^M} = \alpha_j \varepsilon_{jub} \frac{(r_M)_b}{r_M^3} \tag{277}$$

$$= \left(\frac{\boldsymbol{r}_M}{r_M^3} \times \boldsymbol{\alpha} \right)_u . \tag{278}$$

Therefore

$$\boxed{\frac{\partial h^D}{\partial \mu_u^M} = \left(\frac{\boldsymbol{r}_M}{r_M^3} \times \boldsymbol{\alpha} \right)_u .} \tag{279}$$

The Hellmann–Feynman theorem turns either derivative into a current integral. Since

$$\boldsymbol{j}^0(\boldsymbol{r}) = -c\phi_i^{0\dagger}(\boldsymbol{r})\boldsymbol{\alpha}\phi_i^0(\boldsymbol{r}), \tag{280}$$

we have

$$\left\langle \frac{\partial h^D}{\partial \lambda} \right\rangle = \int \phi_i^{0\dagger} \boldsymbol{\alpha} \cdot \frac{\partial \boldsymbol{A}}{\partial \lambda} \phi_i^0 \, \mathrm{d}V \tag{281}$$

$$= -\frac{1}{c} \int \boldsymbol{j}^0 \cdot \frac{\partial \boldsymbol{A}}{\partial \lambda} \, \mathrm{d}V. \tag{282}$$

In the Webinar-9 normalization used in the main text this yields Eqs. (65)–(66) there.

B.2 Step-by-step field-free Gordon current

← Back to the corresponding main-text step

This derivation assumes a local real scalar potential $V_s(\boldsymbol{r})\mathbf{1}_4$. The more general potential-dependent case is treated in Appendix C.

Step 1: start from the Dirac equation. For one occupied four-component spinor,

$$[c\alpha_l \pi_l + \beta c^2 + V_s]\phi_i = E_i \phi_i, \qquad \pi_l = p_l + \frac{A_l}{c}, \qquad p_l = -\mathrm{i}\partial_l. \tag{283}$$

No long-distance approximation is used here.

Step 2: extract an identity for the spinor itself. Move all terms except $\beta c^2 \phi_i$ to the right,

$$\beta c^2 \phi_i = (E_i - V_s - c\alpha_l \pi_l)\phi_i. \tag{284}$$

Multiply from the left by β, use $\beta^2 = \mathbf{1}_4$, and obtain

$$c^2 \phi_i = \beta(E_i - V_s - c\alpha_l \pi_l)\phi_i \tag{285}$$

$$= [\beta(E_i - V_s) - c\beta\alpha_l \pi_l]\phi_i. \tag{286}$$

Hence

$$\boxed{\phi_i = \frac{1}{c^2}[\beta(E_i - V_s) - c\beta\alpha_l \pi_l]\phi_i.} \tag{287}$$

This identity is not a numerical solution for ϕ_i; it is inserted into the current to replace the velocity matrix α_k by derivative and spin-density terms.

Step 3: symmetrize the Dirac current. The current component is

$$j_k = -c\phi_i^\dagger \alpha_k \phi_i. \tag{288}$$

Write it as two identical copies,

$$j_k = -\frac{c}{2}(\phi_i^\dagger \alpha_k \phi_i + \phi_i^\dagger \alpha_k \phi_i). \tag{289}$$

Equation (287) is inserted into the ket of the first copy and its Hermitian conjugate into the bra of the second copy.

Step 4: substitution into the first copy. Using Eq. (287),

$$\phi_i^\dagger \alpha_k \phi_i \longrightarrow \frac{1}{c^2}\phi_i^\dagger \alpha_k[\beta(E_i - V_s) - c\beta\alpha_l\pi_l]\phi_i \tag{290}$$

$$= \frac{E_i}{c^2}\phi_i^\dagger \alpha_k \beta \phi_i - \frac{1}{c^2}\phi_i^\dagger \alpha_k \beta V_s \phi_i - \frac{1}{c}\phi_i^\dagger \alpha_k \beta \alpha_l \pi_l \phi_i. \tag{291}$$

Step 5: Hermitian-conjugate identity for the bra. For real E_i, V_s, with $\alpha_l^\dagger = \alpha_l$ and $\beta^\dagger = \beta$,

$$\phi_i^\dagger = \frac{1}{c^2}[E_i \phi_i^\dagger \beta - (V_s\phi_i)^\dagger \beta - c(\pi_l \phi_i)^\dagger \alpha_l \beta]. \tag{292}$$

Therefore

$$\phi_i^\dagger \alpha_k \phi_i \longrightarrow \frac{E_i}{c^2}\phi_i^\dagger \beta \alpha_k \phi_i - \frac{1}{c^2}(V_s\phi_i)^\dagger \beta \alpha_k \phi_i \tag{293}$$

$$- \frac{1}{c}(\pi_l\phi_i)^\dagger \alpha_l \beta \alpha_k \phi_i. \tag{294}$$

Step 6: collect energy, potential, and kinetic terms. Substitution into Eq. (289) gives

$$j_k = -\frac{E_i}{2c}\phi_i^\dagger(\alpha_k\beta + \beta\alpha_k)\phi_i \tag{295}$$

$$+ \frac{1}{2c}[\phi_i^\dagger \alpha_k \beta V_s \phi_i + (V_s\phi_i)^\dagger \beta \alpha_k \phi_i] \tag{296}$$

$$+ \frac{1}{2}\phi_i^\dagger \alpha_k \beta \alpha_l \pi_l \phi_i + \frac{1}{2}(\pi_l\phi_i)^\dagger \alpha_l \beta \alpha_k \phi_i. \tag{297}$$

The energy term vanishes because

$$\{\alpha_k, \beta\} = 0. \tag{298}$$

For a local real scalar $V_s = v_s(\boldsymbol{r})\mathbf{1}_4$,

$$\frac{1}{2c}[\phi_i^\dagger \alpha_k \beta V_s \phi_i + (V_s\phi_i)^\dagger \beta \alpha_k \phi_i] \tag{299}$$

$$= \frac{v_s}{2c}\phi_i^\dagger(\alpha_k\beta + \beta\alpha_k)\phi_i = 0. \tag{300}$$

This is the precise point at which a scalar potential disappears from the standard Gordon current.

Step 7: expand the kinetic momentum. From

$$\pi_l \phi_i = -\mathrm{i}\partial_l \phi_i + \frac{A_l}{c}\phi_i, \tag{301}$$

we obtain

$$(\pi_l\phi_i)^\dagger = \mathrm{i}\partial_l \phi_i^\dagger + \frac{A_l}{c}\phi_i^\dagger. \tag{302}$$

Equation (297) becomes

$$j_k = \frac{\mathrm{i}}{2}[(\partial_l \phi_i^\dagger)\alpha_l \beta \alpha_k \phi_i - \phi_i^\dagger \alpha_k \beta \alpha_l(\partial_l \phi_i)] \tag{303}$$

$$+ \frac{A_l}{2c}\phi_i^\dagger(\alpha_k\beta\alpha_l + \alpha_l\beta\alpha_k)\phi_i. \tag{304}$$

Step 8: reduce the explicit vector-potential term. Using $\alpha_k\beta = -\beta\alpha_k$ and $\{\alpha_k, \alpha_l\} = 2\delta_{kl}\mathbf{1}_4$,

$$\alpha_k\beta\alpha_l + \alpha_l\beta\alpha_k = -\beta\alpha_k\alpha_l - \beta\alpha_l\alpha_k \tag{305}$$
$$= -\beta\{\alpha_k, \alpha_l\} \tag{306}$$
$$= -2\beta\delta_{kl}. \tag{307}$$

Thus the explicit magnetic-current term is

$$-\frac{A_k}{c}\rho_0^G, \qquad \rho_0^G \equiv \phi_i^\dagger\beta\phi_i. \tag{308}$$

Step 9: split the derivative part into orbital and spin terms. Use

$$\alpha_k\alpha_l = \delta_{kl}\mathbf{1}_4 + \mathrm{i}\varepsilon_{klm}\Sigma_m, \qquad \Sigma_m = \begin{pmatrix}\sigma_m & 0\\ 0 & \sigma_m\end{pmatrix}. \tag{309}$$

The two ordered products are

$$\alpha_l\beta\alpha_k = -\beta\alpha_l\alpha_k = -\delta_{lk}\beta + \mathrm{i}\varepsilon_{klm}\beta\Sigma_m, \tag{310}$$
$$\alpha_k\beta\alpha_l = -\beta\alpha_k\alpha_l = -\delta_{kl}\beta - \mathrm{i}\varepsilon_{klm}\beta\Sigma_m. \tag{311}$$

The δ_{kl} terms in Eq. (304) give

$$(j_{\mathrm{orb}})_k = \frac{\mathrm{i}}{2}[-(\partial_k\phi_i^\dagger)\beta\phi_i + \phi_i^\dagger\beta(\partial_k\phi_i)] \tag{312}$$
$$= -\operatorname{Im}\{\phi_i^\dagger\beta\partial_k\phi_i\}. \tag{313}$$

The Levi–Civita terms give

$$(j_{\mathrm{spin}})_k = -\frac{1}{2}\varepsilon_{klm}[(\partial_l\phi_i^\dagger)\beta\Sigma_m\phi_i + \phi_i^\dagger\beta\Sigma_m(\partial_l\phi_i)] \tag{314}$$
$$= -\frac{1}{2}\varepsilon_{klm}\partial_l(\phi_i^\dagger\beta\Sigma_m\phi_i). \tag{315}$$

Define

$$\rho_m^G \equiv \phi_i^\dagger\beta\Sigma_m\phi_i. \tag{316}$$

Then

$$\boxed{\boldsymbol{j}_{\mathrm{spin}} = -\frac{1}{2}\nabla\times\boldsymbol{\rho}^G.} \tag{317}$$

Combining the orbital, spin, and explicit vector-potential pieces gives

$$\boxed{\boldsymbol{j} = -\operatorname{Im}\{\phi_i^\dagger\beta\nabla\phi_i\} - \frac{1}{2}\nabla\times\boldsymbol{\rho}^G - \frac{1}{c}\boldsymbol{A}\rho_0^G.} \tag{318}$$

At the field-free point, $\boldsymbol{A} = 0$,

$$\boxed{\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} = -\operatorname{Im}\{\phi_i^{0\dagger}\beta\nabla\phi_i^0\} - \frac{1}{2}\nabla\times\boldsymbol{\rho}^{G,0}.} \tag{319}$$

The momentum form used in integrations by parts follows from $\boldsymbol{p} = -\mathrm{i}\nabla$:

$$(\boldsymbol{j}^0_{\mathrm{orb}})_k = -\frac{1}{2}[\phi_i^{0\dagger}p_k\beta\phi_i^0 - (p_k\phi_i^{0\dagger})\beta\phi_i^0]. \tag{320}$$

Step 10: large/small-component content. For

$$\phi_i = \begin{pmatrix}\phi_i^L\\ \phi_i^S\end{pmatrix}, \qquad \beta = \begin{pmatrix}\mathbf{1}_2 & 0\\ 0 & -\mathbf{1}_2\end{pmatrix}, \qquad \Sigma_m = \begin{pmatrix}\sigma_m & 0\\ 0 & \sigma_m\end{pmatrix}, \tag{321}$$

we find

$$\rho_0^G = (\phi_i^L)^\dagger\phi_i^L - (\phi_i^S)^\dagger\phi_i^S, \tag{322}$$
$$\rho_m^G = (\phi_i^L)^\dagger\sigma_m\phi_i^L - (\phi_i^S)^\dagger\sigma_m\phi_i^S. \tag{323}$$

The original Dirac current is instead off-diagonal in large/small space,

$$j_k = -c[(\phi_i^L)^\dagger\sigma_k\phi_i^S + (\phi_i^S)^\dagger\sigma_k\phi_i^L]. \tag{324}$$

The Gordon decomposition is therefore an exact reorganization of the same large–small coupling, not a nonrelativistic approximation.

B.3 Derivation of the two Gordon integration-by-parts identities

← Back to the corresponding main-text step

The entire Gordon decomposition of both g and A^M follows from two integrations by parts. We derive them once for a general real vector field $\boldsymbol{a}(\boldsymbol{r})$.

Orbital-current identity. Define

$$\mathcal{I}_u^{\mathrm{orb}}[\boldsymbol{a}] = \int \varepsilon_{uwk} a_w [\phi^\dagger p_k \beta\phi - (p_k\phi^\dagger)\beta\phi] \,\mathrm{d}V. \tag{325}$$

With $p_k = -\mathrm{i}\partial_k$,

$$\mathcal{I}_u^{\mathrm{orb}} = -\mathrm{i}\int \varepsilon_{uwk} a_w \phi^\dagger \beta \partial_k \phi \,\mathrm{d}V \tag{326}$$

$$+\mathrm{i}\int \varepsilon_{uwk} a_w (\partial_k\phi^\dagger)\beta\phi \,\mathrm{d}V. \tag{327}$$

Integrate the second term by parts:

$$\mathrm{i}\int \varepsilon_{uwk} a_w (\partial_k\phi^\dagger)\beta\phi \,\mathrm{d}V = -\mathrm{i}\int \varepsilon_{uwk} \phi^\dagger \partial_k (a_w\beta\phi) \,\mathrm{d}V \tag{328}$$

$$= -\mathrm{i}\int \varepsilon_{uwk} (\partial_k a_w)\phi^\dagger\beta\phi \,\mathrm{d}V \tag{329}$$

$$-\mathrm{i}\int \varepsilon_{uwk} a_w \phi^\dagger \beta\partial_k\phi \,\mathrm{d}V, \tag{330}$$

where the surface term is zero for the localized states considered here. Consequently

$$\mathcal{I}_u^{\mathrm{orb}}[\boldsymbol{a}] = 2\int \phi^\dagger (\boldsymbol{a}\times\boldsymbol{p})_u \beta\phi \,\mathrm{d}V + \mathrm{i}\int (\nabla\times\boldsymbol{a})_u \phi^\dagger\beta\phi \,\mathrm{d}V. \tag{331}$$

For all geometric fields used in this work,

$$\nabla\times\boldsymbol{r}_G = 0, \qquad \nabla\times\frac{\boldsymbol{r}_M}{r_M^3} = 0, \qquad \nabla\times\boldsymbol{a}^{M,\mathrm{LD}} = 0, \tag{332}$$

so Eq. (331) reduces to

$$\boxed{\mathcal{I}_u^{\mathrm{orb}}[\boldsymbol{a}] = 2\int \phi^\dagger (\boldsymbol{a}\times\boldsymbol{p})_u \beta\phi \,\mathrm{d}V.} \tag{333}$$

This identity generates OZ for $\boldsymbol{a} = \boldsymbol{r}_G$ and PSO for $\boldsymbol{a} = \boldsymbol{r}_M/r_M^3$.

Spin-current identity. Define

$$\mathcal{I}_u^{\mathrm{spin}}[\boldsymbol{a}] = \int [\boldsymbol{a}\times(\nabla\times\boldsymbol{\rho})]_u \,\mathrm{d}V. \tag{334}$$

In components,

$$\mathcal{I}_u^{\mathrm{spin}} = \int \varepsilon_{uwj} a_w \varepsilon_{jmn} \partial_m \rho_n \,\mathrm{d}V \tag{335}$$

$$= \int (\delta_{um}\delta_{wn} - \delta_{un}\delta_{wm}) a_w \partial_m \rho_n \,\mathrm{d}V \tag{336}$$

$$= \int (a_n \partial_u \rho_n - a_m \partial_m \rho_u) \,\mathrm{d}V. \tag{337}$$

Integrating the two terms by parts separately,

$$\int a_n \partial_u \rho_n \,\mathrm{d}V = -\int (\partial_u a_n)\rho_n \,\mathrm{d}V, \tag{338}$$

$$-\int a_m \partial_m \rho_u \,\mathrm{d}V = \int (\partial_m a_m)\rho_u \,\mathrm{d}V, \tag{339}$$

again neglecting the surface terms of a localized state. Hence

$$\boxed{\mathcal{I}_u^{\mathrm{spin}}[\boldsymbol{a}] = \int [(\partial_m a_m)\rho_u - (\partial_u a_n)\rho_n]\,\mathrm{d}V.} \tag{340}$$

This one formula generates SZ for $\boldsymbol{a} = \boldsymbol{r}_G$, FC+SD for the exact nuclear field, and the transferred spin-Zeeman term for the smooth long-distance field.

B.4 Exact local FC and SD from the Gordon spin current

← Back to the corresponding main-text step

For the exact nuclear field

$$\boldsymbol{a}^M = \frac{\boldsymbol{r}_M}{r_M^3}, \tag{341}$$

the orbital-current identity immediately gives the PSO contribution,

$$A_{uv}^{M,\mathrm{PSO}} = \frac{\gamma^M}{2\pi c}\int \phi_i^{0\dagger}(J_v)2\left[\frac{\boldsymbol{r}_M \times \boldsymbol{p}}{r_M^3}\right]_u \beta\phi_i^0(J_v)\,\mathrm{d}V, \tag{342}$$

in the Webinar-9 normalization of the main text.

The spin contribution must first be kept unsplit:

$$A_{uv}^{M,\mathrm{spin}} = \frac{\gamma^M}{2\pi c}\int [(\partial_w a_w^M)\rho_u^{G,0} - (\partial_u a_w^M)\rho_w^{G,0}]\,\mathrm{d}V. \tag{343}$$

The distributional derivative is

$$\partial_u a_w^M = \mathrm{PV}\left(\frac{\delta_{uw} r_M^2 - 3(r_M)_u(r_M)_w}{r_M^5}\right) + \frac{4\pi}{3}\delta_{uw}\delta^{(3)}(\boldsymbol{r}_M), \tag{344}$$

and its trace is

$$\partial_w a_w^M = 4\pi\delta^{(3)}(\boldsymbol{r}_M). \tag{345}$$

Substitute both into Eq. (343):

$$(\partial_w a_w^M)\rho_u^{G,0} - (\partial_u a_w^M)\rho_w^{G,0} \tag{346}$$

$$= 4\pi\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0} - \frac{4\pi}{3}\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0} \tag{347}$$

$$+ \mathrm{PV}\left[\frac{3(r_M)_u(r_M)_w - \delta_{uw} r_M^2}{r_M^5}\right]\rho_w^{G,0} \tag{348}$$

$$= \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0} + \mathrm{PV}\left[\frac{3(r_M)_u(r_M)_w - \delta_{uw} r_M^2}{r_M^5}\right]\rho_w^{G,0}. \tag{349}$$

Therefore

$$A_{uv}^{M,\mathrm{FC}} = \frac{\gamma^M}{2\pi c}\int \frac{8\pi}{3}\delta^{(3)}(\boldsymbol{r}_M)\rho_u^{G,0}\,\mathrm{d}V, \tag{350}$$

$$A_{uv}^{M,\mathrm{SD}} = \frac{\gamma^M}{2\pi c}\int \frac{3(r_M)_u(r_M)_w - \delta_{uw} r_M^2}{r_M^5}\rho_w^{G,0}\,\mathrm{d}V. \tag{351}$$

The same $8\pi/3$ coefficient has therefore been obtained by a completely different route from the RKB calculation.

B.5 Derivation of the long-distance gauge identity

← Back to the corresponding main-text step

After the long-distance expansion,

$$\boldsymbol{a}^{M,\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R})\boldsymbol{r}_G. \tag{352}$$

We want to decompose the vector-potential derivative $\boldsymbol{e}_u \times \boldsymbol{a}^{M,\mathrm{LD}}$ into a homogeneous-field part plus a gradient. The claim is

$$\boxed{\boldsymbol{e}_u \times \boldsymbol{a}^{M,\mathrm{LD}} = \frac{1}{2}T_{ku}(\boldsymbol{R})(\boldsymbol{e}_k \times \boldsymbol{r}_G) + \nabla\chi_u,} \tag{353}$$

with

$$\chi_u = \frac{(\boldsymbol{R}\times\boldsymbol{r}_G)_u}{R^3} - \frac{3}{2R^5}(\boldsymbol{R}\cdot\boldsymbol{r}_G)(\boldsymbol{R}\times\boldsymbol{r}_G)_u. \tag{354}$$

To verify it componentwise, write the q-component of the left-hand side as

$$[\boldsymbol{e}_u \times \boldsymbol{a}^{M,\mathrm{LD}}]_q = \varepsilon_{quw}\left[\frac{R_w}{R^3} - T_{wn}(r_G)_n\right]. \tag{355}$$

The uniform-field term on the right is

$$\frac{1}{2}T_{ku}(\boldsymbol{e}_k \times \boldsymbol{r}_G)_q = \frac{1}{2}\left(\frac{3R_kR_u}{R^5} - \frac{\delta_{ku}}{R^3}\right)\varepsilon_{qkn}(r_G)_n. \tag{356}$$

Differentiate Eq. (354). Since

$$\partial_q(\boldsymbol{R}\cdot\boldsymbol{r}_G) = R_q, \qquad \partial_q(\boldsymbol{R}\times\boldsymbol{r}_G)_u = \varepsilon_{uaq}R_a, \tag{357}$$

we get

$$\partial_q\chi_u = \frac{\varepsilon_{uaq}R_a}{R^3} \tag{358}$$

$$- \frac{3}{2R^5}\left[R_q(\boldsymbol{R}\times\boldsymbol{r}_G)_u + (\boldsymbol{R}\cdot\boldsymbol{r}_G)\varepsilon_{uaq}R_a\right]. \tag{359}$$

Adding Eqs. (356) and (359), expanding $(\boldsymbol{R}\times\boldsymbol{r}_G)_u = \varepsilon_{uab}R_a(r_G)_b$, and using the two-epsilon contraction reduces the result to Eq. (355). Thus Eq. (353) is an exact identity for the retained first-order field; no additional approximation is introduced here.

At the Hamiltonian level,

$$\left(\frac{\partial h^D}{\partial \mu_u^M}\right)_{\mathrm{LD}} = T_{ku}(\boldsymbol{R})\frac{\partial h^D}{\partial B_k} + \boldsymbol{\alpha}\cdot\nabla\chi_u, \tag{360}$$

with the overall normalization understood in the Webinar convention. When the same origin is used in the RKB and Gordon routes, $\boldsymbol{r}_O = \boldsymbol{r}_G$, their gauge functions satisfy

$$f_u = \alpha^2\chi_u. \tag{361}$$

Therefore the RKB commutator and the Gordon gradient term are two representations of the same gauge remainder.

B.6 Why the Gordon gauge-gradient term vanishes

← Back to the corresponding main-text step

Define

$$\mathcal{I}_u^G = \int \boldsymbol{j}^0\cdot\nabla\chi_u\,\mathrm{d}V. \tag{362}$$

Using

$$\nabla\cdot(\chi_u\boldsymbol{j}^0) = \boldsymbol{j}^0\cdot\nabla\chi_u + \chi_u\nabla\cdot\boldsymbol{j}^0, \tag{363}$$

we obtain

$$\mathcal{I}_u^G = \int \nabla\cdot(\chi_u\boldsymbol{j}^0)\,\mathrm{d}V - \int \chi_u\nabla\cdot\boldsymbol{j}^0\,\mathrm{d}V \tag{364}$$

$$= \oint_{\partial\Omega}\chi_u\boldsymbol{j}^0\cdot\mathrm{d}\boldsymbol{S} - \int\chi_u\nabla\cdot\boldsymbol{j}^0\,\mathrm{d}V. \tag{365}$$

For a localized molecular bound state the current decays sufficiently fast that the surface term vanishes as the boundary is taken to infinity. For a stationary state,

$$\partial_t\rho + \nabla\cdot\boldsymbol{j}^0 = 0, \qquad \partial_t\rho = 0, \tag{366}$$

so

$$\nabla \cdot \boldsymbol{j}^0 = 0 \implies \boxed{\mathcal{I}_u^G = 0.} \tag{367}$$

This is the current-density equivalent of $\langle \mathrm{i}[h^{D,(0)}, f_u] \rangle = 0$.

In the simplified Webinar current,

$$\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}}, \qquad \boldsymbol{j}^0_{\mathrm{spin}} = -\frac{1}{2}\nabla \times \boldsymbol{\rho}^{G,0}, \tag{368}$$

and therefore

$$\nabla \cdot \boldsymbol{j}^0_{\mathrm{spin}} = -\frac{1}{2}\nabla \cdot (\nabla \times \boldsymbol{\rho}^{G,0}) = 0 \tag{369}$$

identically. Stationary continuity then also implies

$$\nabla \cdot \boldsymbol{j}^0_{\mathrm{orb}} = 0, \tag{370}$$

so the orbital gradient term may vanish separately in this simplified decomposition.

The potential-dependent DKS current of Appendix C is subtler:

$$\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} + \boldsymbol{j}^0_{\mathrm{xc}}. \tag{371}$$

Only the spin current is identically divergence-free. Thus continuity implies

$$\nabla \cdot (\boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{xc}}) = 0, \tag{372}$$

not necessarily the separate vanishing of the two divergences. This is why the Debora route requires a cancellation between two gauge residuals rather than setting each one to zero independently.

C Detailed potential-dependent DKS–Gordon derivations

C.1 Derivation of Debora's potential current and the xc term

← Back to the corresponding main-text step

The potential-dependent derivation starts from the rest-energy-shifted DKS equation

$$[c\alpha_l \pi_l + (\beta - \mathbf{1}_4)c^2 + V]\phi_i = \varepsilon_i \phi_i, \qquad \pi_l = p_l + \frac{A_l}{c}. \tag{373}$$

The asymmetric-looking large/small diagonal blocks are caused only by subtraction of $c^2\mathbf{1}_4$; they are not the origin of the extra potential current.

Step 1: isolate the spinor. Expand

$$(\beta - \mathbf{1}_4)c^2 = \beta c^2 - c^2\mathbf{1}_4 \tag{374}$$

and move all terms except $\beta c^2 \phi_i$ to the right:

$$\beta c^2 \phi_i = [\varepsilon_i + c^2 - V - c\alpha_l \pi_l]\phi_i. \tag{375}$$

Define the scalar energy combination

$$\Lambda_i \equiv \varepsilon_i + c^2. \tag{376}$$

Multiplication by β gives

$$\boxed{\phi_i = \frac{1}{c^2}[\beta(\Lambda_i - V) - c\beta\alpha_l \pi_l]\phi_i.} \tag{377}$$

The precise numerical zero hidden in Λ_i will disappear from the current because it multiplies $\{\alpha_k, \beta\}$.

Step 2: symmetrize the current and substitute Eq. (377). Start from

$$j_k = -\frac{c}{2}(\phi_i^\dagger \alpha_k \phi_i + \phi_i^\dagger \alpha_k \phi_i). \tag{378}$$

Inserting Eq. (377) in the ket of the first copy gives

$$\phi_i^\dagger \alpha_k \phi_i \longrightarrow \frac{\Lambda_i}{c^2}\phi_i^\dagger \alpha_k \beta \phi_i - \frac{1}{c^2}\phi_i^\dagger \alpha_k \beta (V\phi_i) \tag{379}$$

$$- \frac{1}{c}\phi_i^\dagger \alpha_k \beta \alpha_l \pi_l \phi_i. \tag{380}$$

The Hermitian conjugate of Eq. (377) is

$$\phi_i^\dagger = \frac{1}{c^2}[\Lambda_i \phi_i^\dagger \beta - (V\phi_i)^\dagger \beta - c(\pi_l \phi_i)^\dagger \alpha_l \beta], \tag{381}$$

so the second copy becomes

$$\phi_i^\dagger \alpha_k \phi_i \longrightarrow \frac{\Lambda_i}{c^2}\phi_i^\dagger \beta \alpha_k \phi_i - \frac{1}{c^2}(V\phi_i)^\dagger \beta \alpha_k \phi_i \tag{382}$$

$$- \frac{1}{c}(\pi_l \phi_i)^\dagger \alpha_l \beta \alpha_k \phi_i. \tag{383}$$

After multiplication by $-c/2$, the current separates into

$$j_k = -\frac{\Lambda_i}{2c}\phi_i^\dagger (\alpha_k \beta + \beta \alpha_k)\phi_i \tag{384}$$

$$+ \frac{1}{2c}[\phi_i^\dagger \alpha_k \beta (V\phi_i) + (V\phi_i)^\dagger \beta \alpha_k \phi_i] \tag{385}$$

$$+ \frac{1}{2}\phi_i^\dagger \alpha_k \beta \alpha_l \pi_l \phi_i + \frac{1}{2}(\pi_l \phi_i)^\dagger \alpha_l \beta \alpha_k \phi_i. \tag{386}$$

The first line vanishes identically because $\{\alpha_k, \beta\} = 0$. This proves explicitly that the rest-energy shift only changes the energy zero.

Step 3: the kinetic term is the same Gordon algebra as before. Expand

$$\pi_l \phi_i = -\mathrm{i}\partial_l \phi_i + \frac{A_l}{c}\phi_i, \qquad (\pi_l \phi_i)^\dagger = \mathrm{i}\partial_l \phi_i^\dagger + \frac{A_l}{c}\phi_i^\dagger. \tag{387}$$

The last line of Eq. (386) then produces exactly the orbital, spin, and explicit vector-potential pieces already derived in Appendix B:

$$\boldsymbol{j}_{\mathrm{kin}} = \boldsymbol{j}_{\mathrm{orb}} + \boldsymbol{j}_{\mathrm{spin}} - \frac{1}{c}\boldsymbol{A}\rho_0^G. \tag{388}$$

The only genuinely new object is therefore the unsimplified potential term

$$\boxed{(j_V)_k = \frac{1}{2c}[\phi_i^\dagger \alpha_k \beta (V\phi_i) + (V\phi_i)^\dagger \beta \alpha_k \phi_i].} \tag{389}$$

At zero external vector potential,

$$\boxed{\boldsymbol{j}^0 = \boldsymbol{j}^0_{\mathrm{orb}} + \boldsymbol{j}^0_{\mathrm{spin}} + \boldsymbol{j}^0_V.} \tag{390}$$

Step 4: a local scalar potential gives no extra current. If

$$V_s = v(\boldsymbol{r})\mathbf{1}_4, \tag{391}$$

then $(V_s\phi_i)^\dagger = \phi_i^\dagger v$ and Eq. (389) becomes

$$(j_{V_s})_k = \frac{v}{2c}\phi_i^\dagger (\alpha_k \beta + \beta \alpha_k)\phi_i = 0. \tag{392}$$

Thus nuclear/Coulomb scalar pieces and scalar xc pieces do not create an additional Gordon current in this local form.

Step 5: the noncollinear xc potential survives. Take

$$V^{\mathrm{xc}} = F_l \Sigma_l, \tag{393}$$

with real local coefficients $F_l(\boldsymbol{r})$. Equation (389) becomes

$$(j_{\mathrm{xc}})_k = \frac{F_l}{2c} \phi_i^\dagger (\alpha_k \beta \Sigma_l + \Sigma_l \beta \alpha_k) \phi_i. \tag{394}$$

The required Dirac-matrix identity is

$$\alpha_k \beta \Sigma_l + \Sigma_l \beta \alpha_k = -2\mathrm{i}\beta \varepsilon_{klm} \alpha_m. \tag{395}$$

Hence

$$(j_{\mathrm{xc}})_k = -\frac{\mathrm{i}}{c} \varepsilon_{klm} F_l \phi_i^\dagger \beta \alpha_m \phi_i. \tag{396}$$

Defining

$$Q_m \equiv \phi_i^\dagger \beta \alpha_m \phi_i, \tag{397}$$

we obtain

$$\boxed{\boldsymbol{j}_{\mathrm{xc}} = -\frac{\mathrm{i}}{c} \boldsymbol{F} \times \boldsymbol{Q},} \tag{398}$$

which is Eq. (101).

The corresponding xc contribution to a nuclear magnetic perturbation is easily written at the integrand level. Using

$$(\boldsymbol{F} \times \boldsymbol{Q}) \cdot (\boldsymbol{e}_u \times \boldsymbol{a}) = F_u (\boldsymbol{a} \cdot \boldsymbol{Q}) - (\boldsymbol{a} \cdot \boldsymbol{F}) Q_u, \tag{399}$$

one gets the characteristic combination

$$F_u \, \phi_i^\dagger \beta (\boldsymbol{a} \cdot \boldsymbol{\alpha}) \phi_i - (\boldsymbol{a} \cdot \boldsymbol{F}) \phi_i^\dagger \beta \alpha_u \phi_i, \tag{400}$$

with the overall property prefactor fixed by the HFC convention used in the main text. This is the local xc HFC sector that is mapped in the next subsection.

C.2 Long-distance reduction and cancellation of Debora's gauge residuals

← Back to the corresponding main-text step

The retained nuclear field and its derivatives are the same as in the scalar Gordon route,

$$\boldsymbol{a}^{M,\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} - \boldsymbol{T}(\boldsymbol{R}) \boldsymbol{r}_G, \tag{401}$$

$$\partial_u a_w^{M,\mathrm{LD}} = -T_{wu}, \qquad \partial_w a_w^{M,\mathrm{LD}} = 0, \qquad \nabla \times \boldsymbol{a}^{M,\mathrm{LD}} = 0. \tag{402}$$

The same gauge identity holds,

$$\boldsymbol{e}_u \times \boldsymbol{a}^{M,\mathrm{LD}} = \frac{1}{2} T_{ku} (\boldsymbol{e}_k \times \boldsymbol{r}_G) + \nabla \chi_u. \tag{403}$$

General partial-current mapping. Let X denote one current sector. In the normalization used for Debora's decomposition in the main text, inserting Eq. (403) into the HFC current integral gives

$$A_{uv}^{M,X,\mathrm{LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^X + \Delta_{uv}^X, \ \Delta_{uv}^X \qquad = -\frac{\gamma^M}{2c} \int \boldsymbol{j}_X^0 \cdot \nabla \chi_u \, \mathrm{d}V, \tag{404}$$

which is Eq. (105). The first term is the homogeneous-field response of that current sector; the second is its gauge-gradient residual.

Orbital sector. The orbital current generates PSO in the exact HFC tensor and OZ in the g-tensor. Therefore

$$\boxed{A_{uv}^{M,\mathrm{PSO,LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^{\mathrm{OZ}} + \Delta_{uv}^{\mathrm{orb}}.} \tag{405}$$

Unlike the simplified Webinar decomposition, Δ^{orb} must not be set to zero by itself: in the noncollinear DKS current the orbital sector is not generally separately conserved.

Spin sector: take the long-distance limit before the FC/SD split. The unsplit Gordon spin expression contains

$$(\partial_w a_w^M)\rho_u^{G,0} - (\partial_u a_w^M)\rho_w^{G,0}. \tag{406}$$

Insert Eq. (402):

$$(\partial_w a_w^{M,\mathrm{LD}})\rho_u^{G,0} - (\partial_u a_w^{M,\mathrm{LD}})\rho_w^{G,0} = 0 - (-T_{wu})\rho_w^{G,0} \tag{407}$$

$$= T_{wu}\rho_w^{G,0}. \tag{408}$$

The retained field is smooth, so no distributional derivative is generated. Consequently

$$\boxed{A_{uv}^{M,\mathrm{FC,LD}} = 0,} \tag{409}$$

and the regular spin part becomes

$$\boxed{A_{uv}^{M,\mathrm{SD,LD}} = \frac{\gamma^M}{2c} T_{wu} g_{wv}^{\mathrm{SZ}}.} \tag{410}$$

Equivalently, the spin-current gauge residual vanishes separately because

$$\nabla \cdot \boldsymbol{j}_{\mathrm{spin}}^0 = -\frac{1}{2}\nabla \cdot (\nabla \times \boldsymbol{\rho}^{G,0}) = 0. \tag{411}$$

Exchange–correlation sector. The xc HFC term can be written as a current integral involving $\boldsymbol{j}_{\mathrm{xc}}^0$. Applying Eq. (404) gives

$$\boxed{A_{uv}^{M,\mathrm{xc,LD}} = \frac{\gamma^M}{2c} T_{ku} g_{kv}^{\mathrm{xc}} + \Delta_{uv}^{\mathrm{xc}},} \tag{412}$$

with

$$\Delta_{uv}^{\mathrm{xc}} = -\frac{\gamma^M}{2c} \int \boldsymbol{j}_{\mathrm{xc}}^0 \cdot \nabla \chi_u \, \mathrm{d}V. \tag{413}$$

Thus the xc term does not disappear in the long-distance limit: it becomes the transferred xc part of the g-tensor plus a gauge residual.

Why the two residuals cancel only in the sum. The full field-free DKS current is

$$\boldsymbol{j}^0 = \boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{spin}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0. \tag{414}$$

For a stationary state,

$$\nabla \cdot \boldsymbol{j}^0 = 0. \tag{415}$$

Because $\nabla \cdot \boldsymbol{j}_{\mathrm{spin}}^0 = 0$ identically,

$$\boxed{\nabla \cdot (\boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0) = 0.} \tag{416}$$

Now add the two residuals,

$$\Delta_{uv}^{\mathrm{orb}} + \Delta_{uv}^{\mathrm{xc}} = -\frac{\gamma^M}{2c} \int (\boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0) \cdot \nabla \chi_u \, \mathrm{d}V. \tag{417}$$

Integrating by parts,

$$\int (\boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0) \cdot \nabla \chi_u \, \mathrm{d}V = \oint \chi_u (\boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0) \cdot \mathrm{d}\boldsymbol{S} \tag{418}$$

$$- \int \chi_u \nabla \cdot (\boldsymbol{j}_{\mathrm{orb}}^0 + \boldsymbol{j}_{\mathrm{xc}}^0) \, \mathrm{d}V. \tag{419}$$

The surface term vanishes for a localized molecular state and the volume term vanishes by Eq. (416). Therefore

$$\boxed{\Delta^{\mathrm{orb}}_{uv} + \Delta^{\mathrm{xc}}_{uv} = 0.} \tag{420}$$

The important point is that this is a cancellation in the conserved sum; forcing $\Delta^{\mathrm{orb}} = 0$ and $\Delta^{\mathrm{xc}} = 0$ separately is not justified in general.

Adding all long-distance sectors,

$$A^{M,\mathrm{LD}}_{uv} = \frac{\gamma^M}{2c} T_{ku}(g^{\mathrm{OZ}}_{kv} + g^{\mathrm{SZ}}_{kv} + g^{\mathrm{xc}}_{kv}) \tag{421}$$

$$+ \Delta^{\mathrm{orb}}_{uv} + \Delta^{\mathrm{xc}}_{uv} \tag{422}$$

$$= \boxed{\frac{\gamma^M}{2c} T_{ku}(\boldsymbol{R}) g_{kv}}, \tag{423}$$

which is Eq. (111). This shows explicitly why allowing a noncollinear xc current changes the internal decomposition but not the final long-distance transfer relation.

D Accuracy and interpretation of the approximation

The retained Taylor field is

$$\boldsymbol{a}^{\mathrm{LD}} = \frac{\boldsymbol{R}}{R^3} + \frac{\boldsymbol{r}_O}{R^3} - 3\frac{(\boldsymbol{R}\cdot\boldsymbol{r}_O)\boldsymbol{R}}{R^5}. \tag{424}$$

The next omitted terms are quadratic in the electronic displacement,

$$\boldsymbol{a}^K - \boldsymbol{a}^{\mathrm{LD}} = \mathcal{O}\left(\frac{r_O^2}{R^4}\right). \tag{425}$$

A spatial derivative lowers the length dimension of the omitted contribution, so the corresponding smooth corrections entering the lower RKB/Gordon derivative sector are of order

$$\mathcal{O}\left(\frac{r_O}{R^4}\right) \tag{426}$$

in the local electronic region.

Two different statements must be kept separate.

1. The *finite-order long-distance operator* contains no Fermi-contact distribution, because every retained Taylor coefficient is analytic in $\boldsymbol{r}_O$ around O.
2. The *exact unexpanded molecular matrix element* can contain a nonzero contact contribution if the electronic wavefunction has non-negligible amplitude at the distant nucleus K.

Thus $\mathrm{FC}^{\mathrm{LD}} = 0$ is an operator statement within the localized-centre/no-overlap approximation; it is not the assertion that the exact density at K is mathematically zero.

No finite Taylor order about O can reconstruct $\delta^{(3)}(\boldsymbol{r}_K)$, because that distribution is supported at the singular point $\boldsymbol{r}_K = 0$, outside the analytic neighbourhood in which the expansion is used. Recovering the exact contact contribution requires returning to the unexpanded kernel $\boldsymbol{r}_K/r_K^3$ and evaluating the actual electronic density at K.

Finally, the order of operations matters. The magnetic derivative is first formed from the field-dependent Hamiltonian, the perturbation is then set to zero, and only afterwards is the distant nuclear kernel expanded. Expanding an undifferentiated Hamiltonian too early can remove or reshuffle perturbation-dependent terms and is not equivalent to the procedure used in the derivations above.